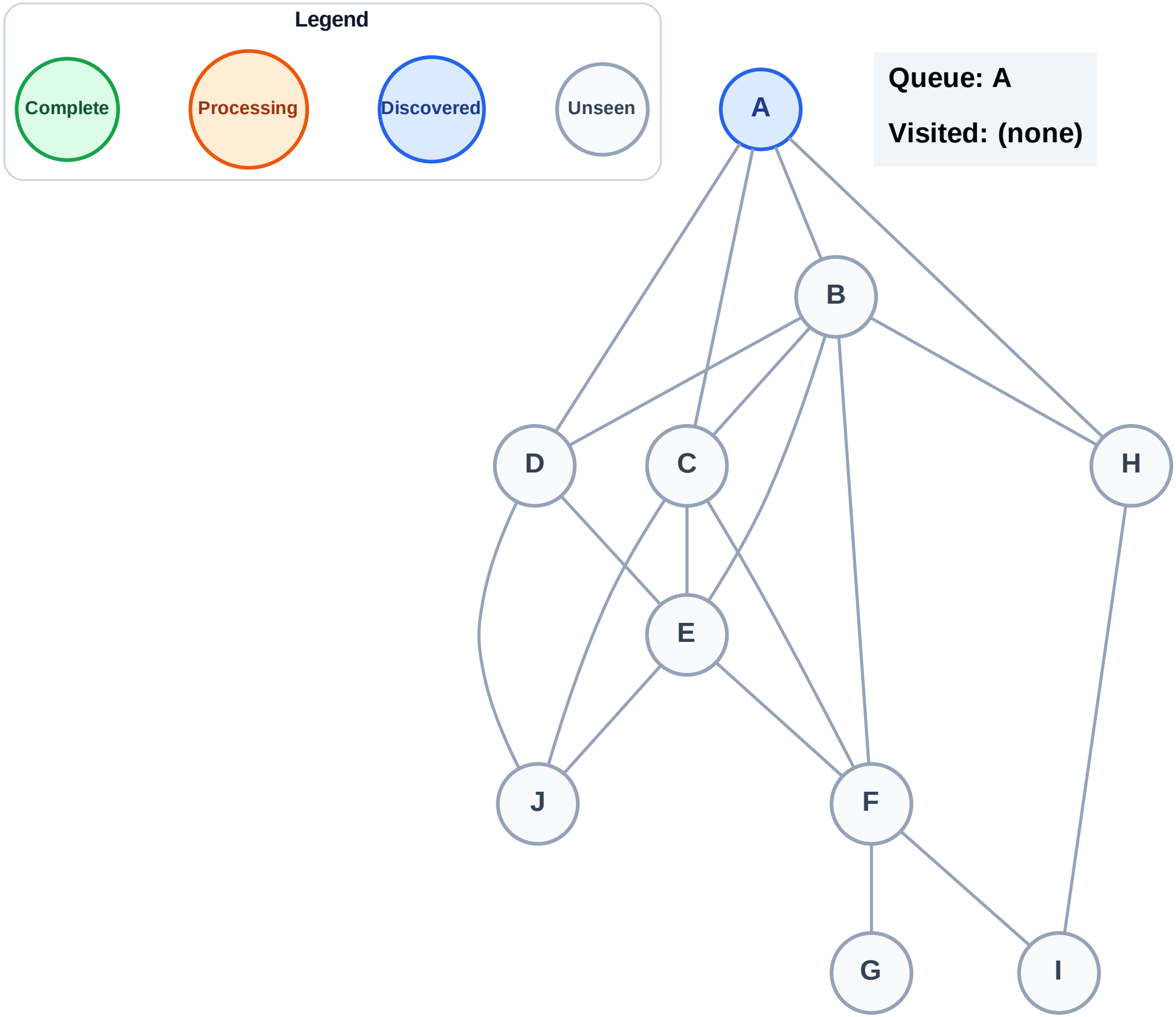
BFS Traversal

Step 1: Start at A

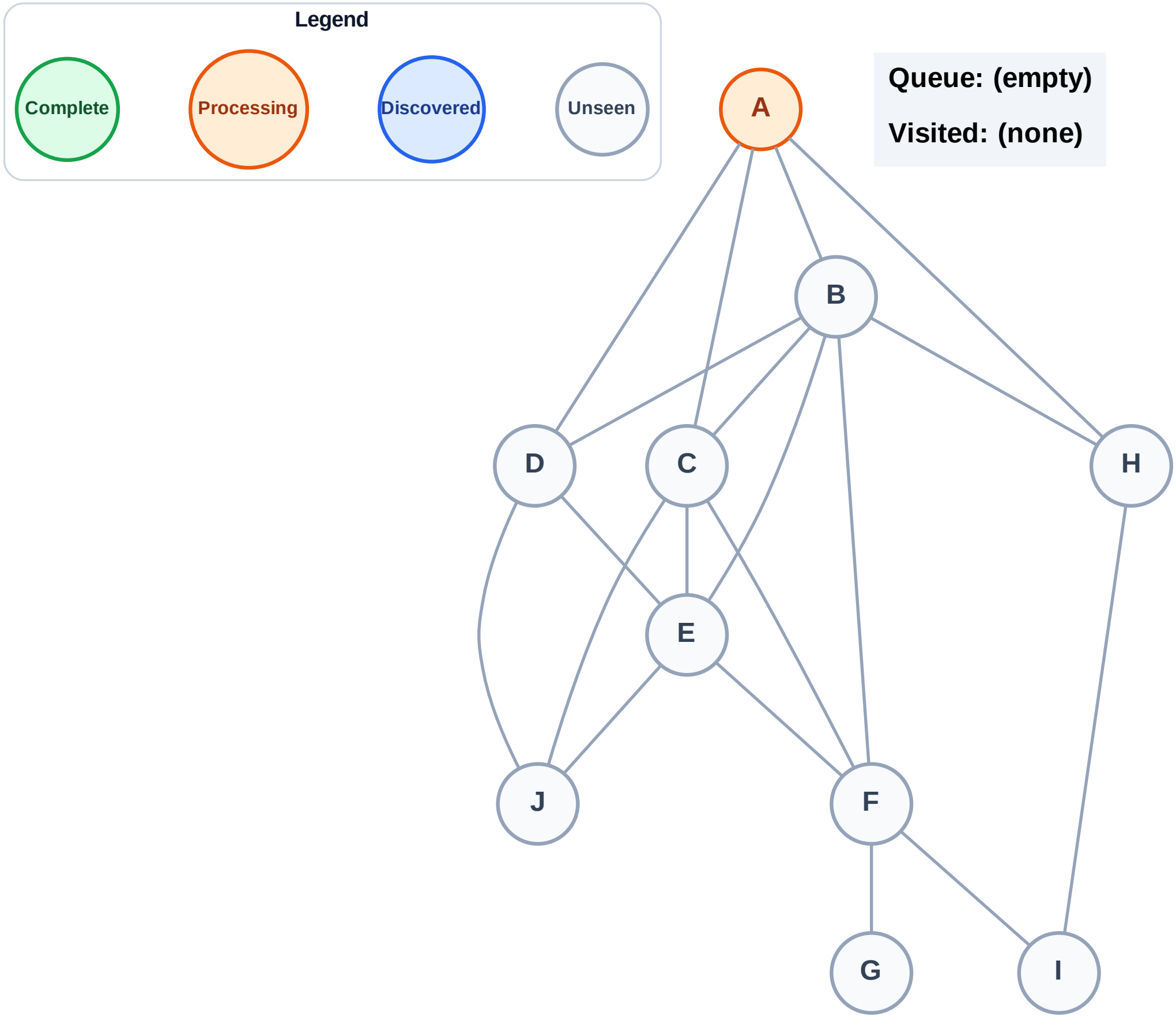


Queue: A

Visited: (none)

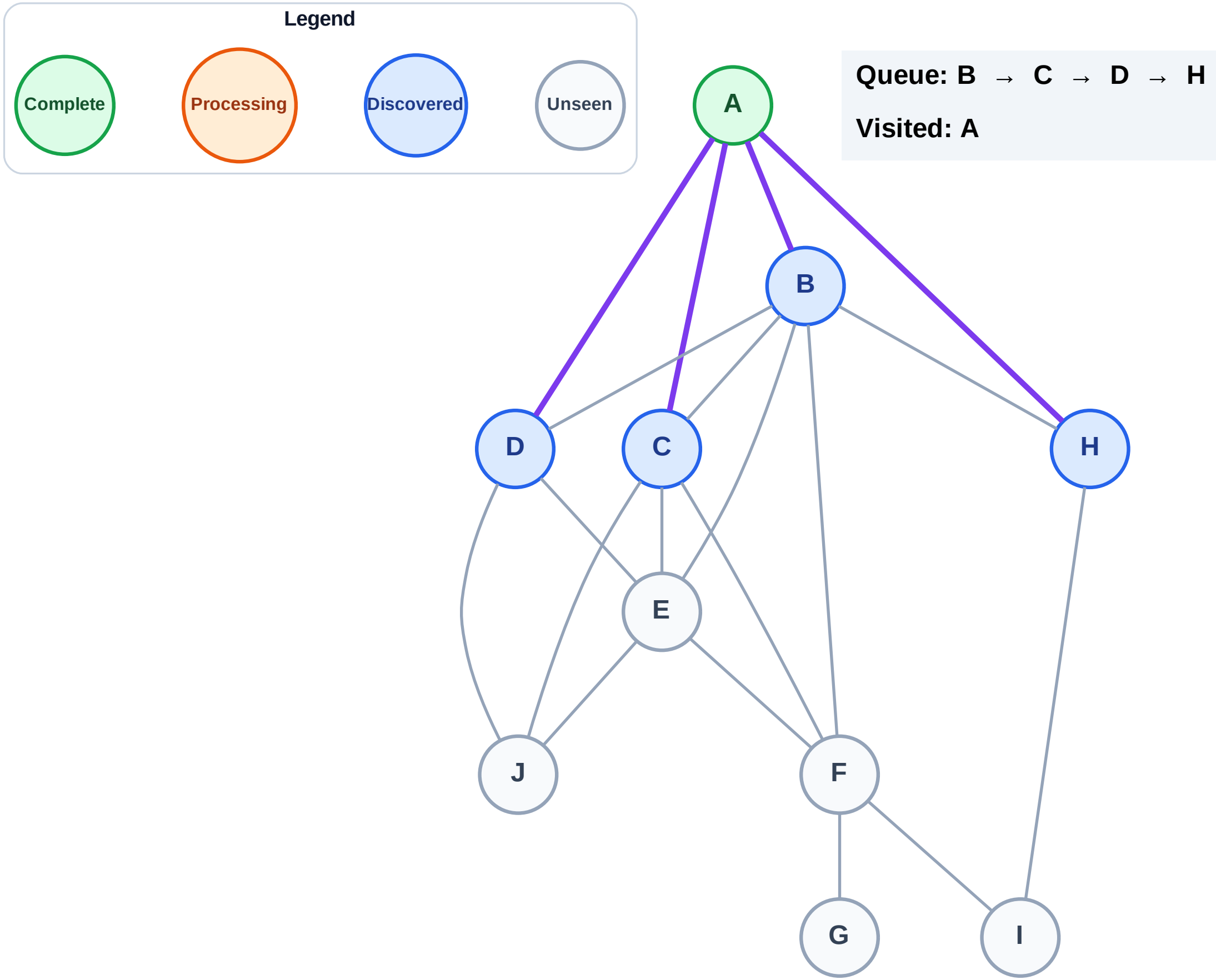
BFS Traversal

Step 2: Process node A



BFS Traversal

Step 3: Discover B, C, D, H from A

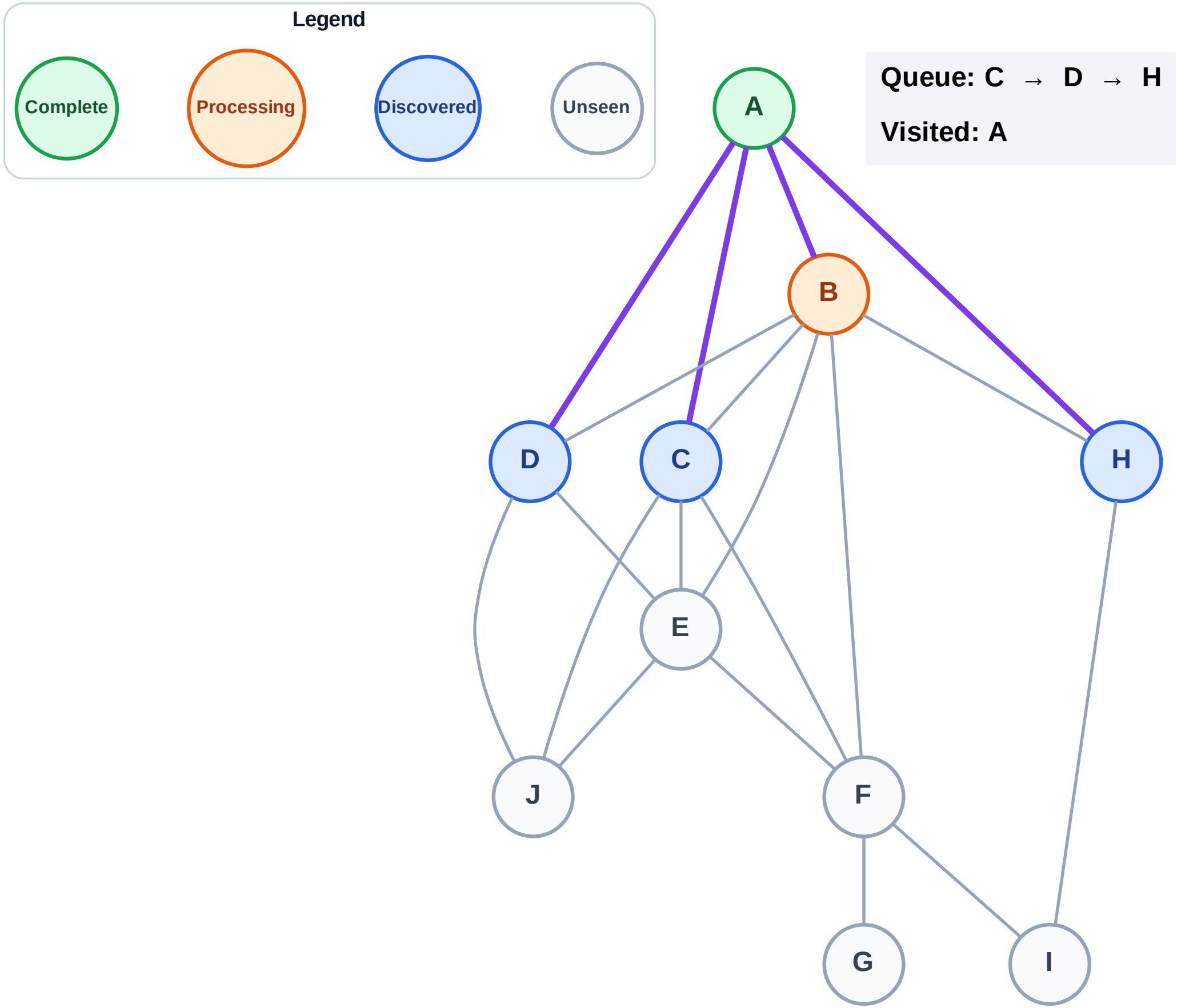


Queue: B → C → D → H

Visited: A

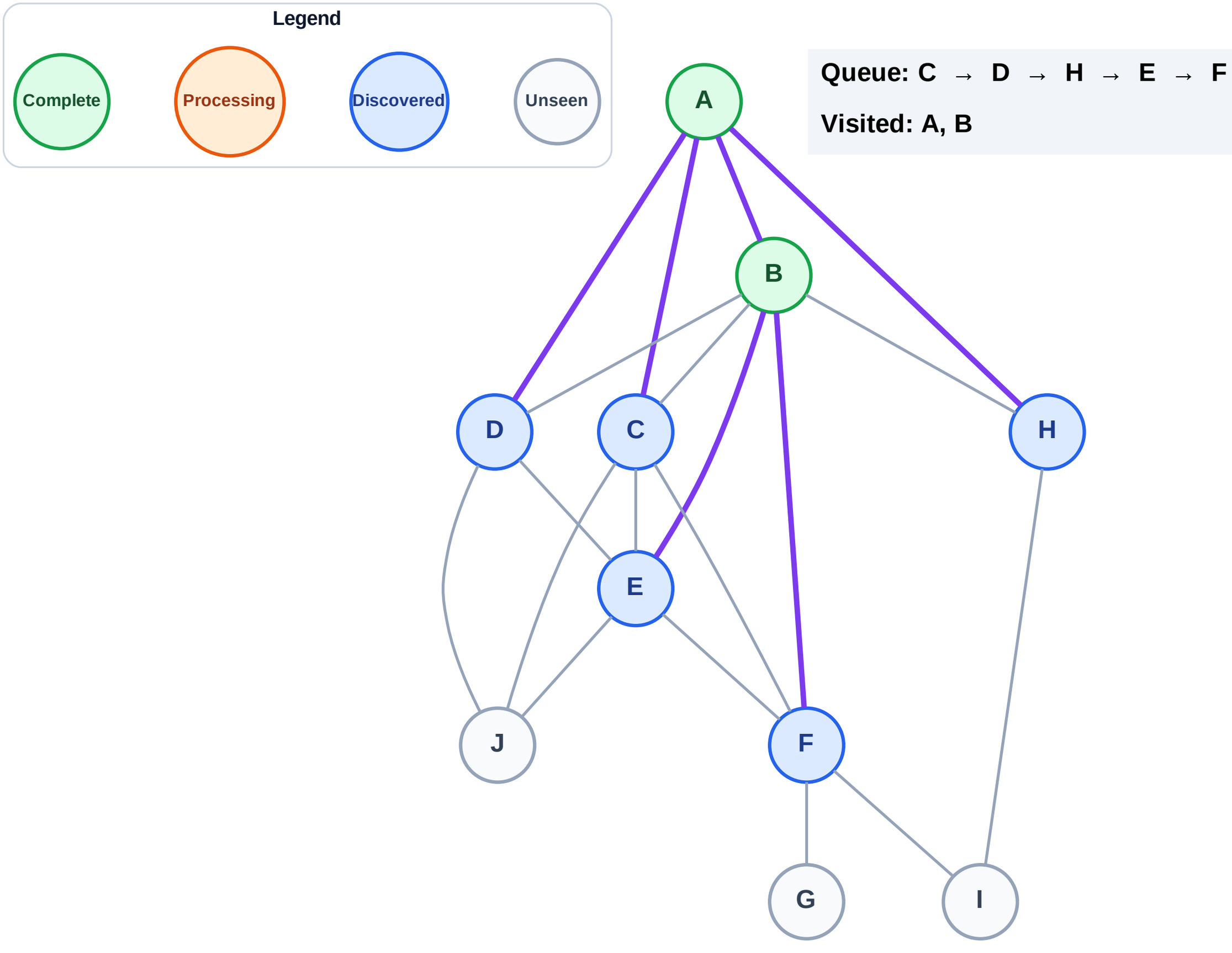
BFS Traversal

Step 4: Process node B



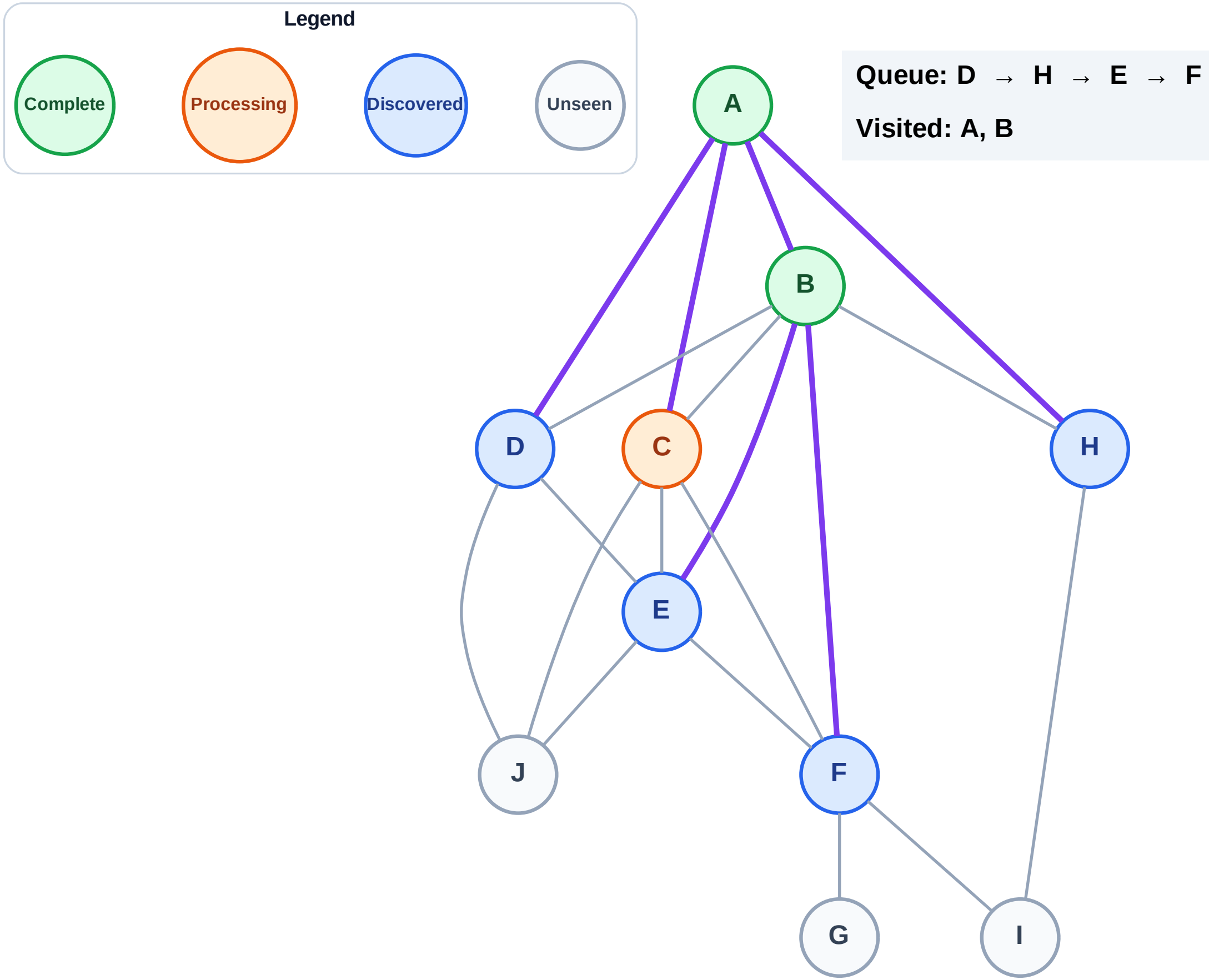
BFS Traversal

Step 5: Discover E, F from B



BFS Traversal

Step 6: Process node C

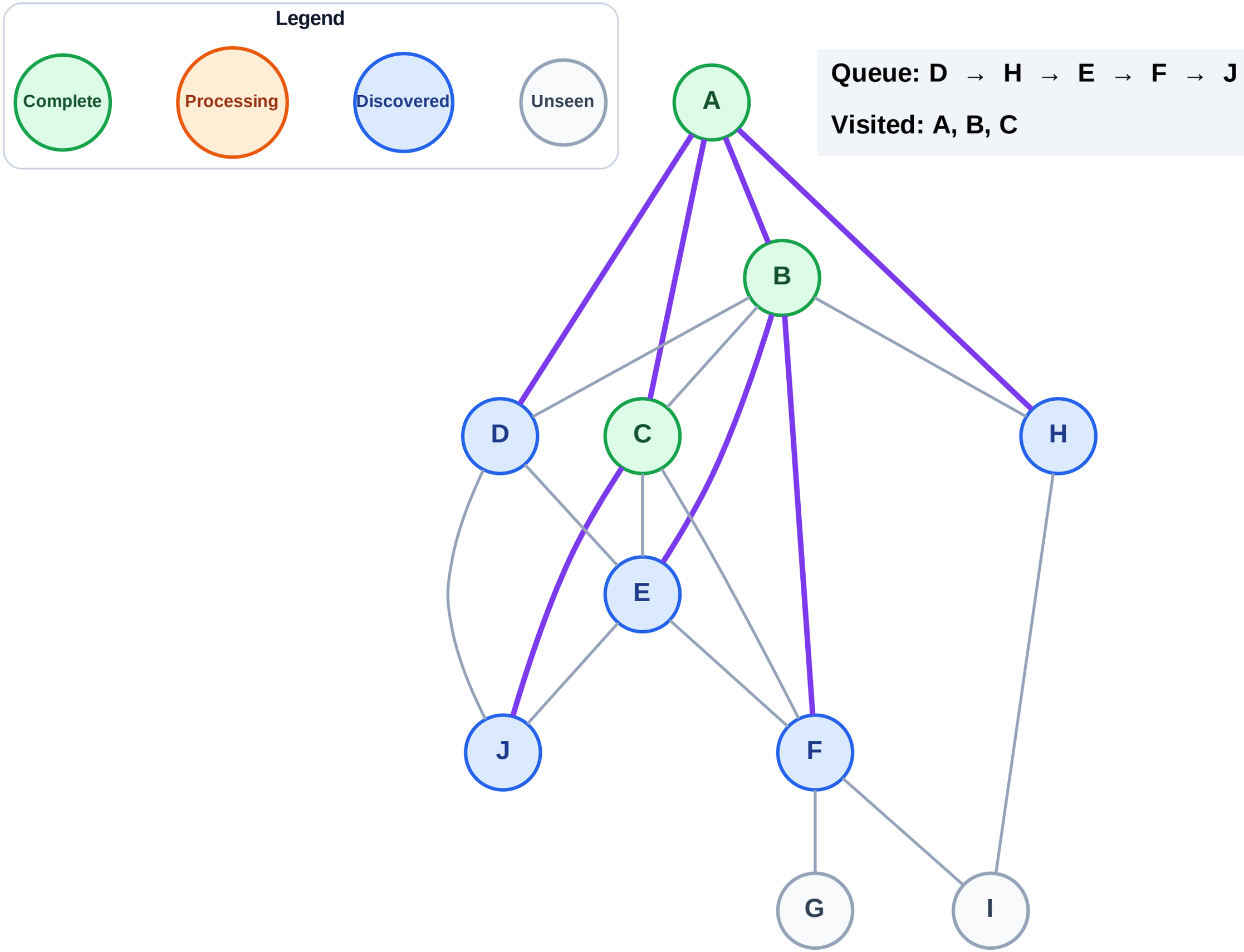


Queue: D → H → E → F

Visited: A, B

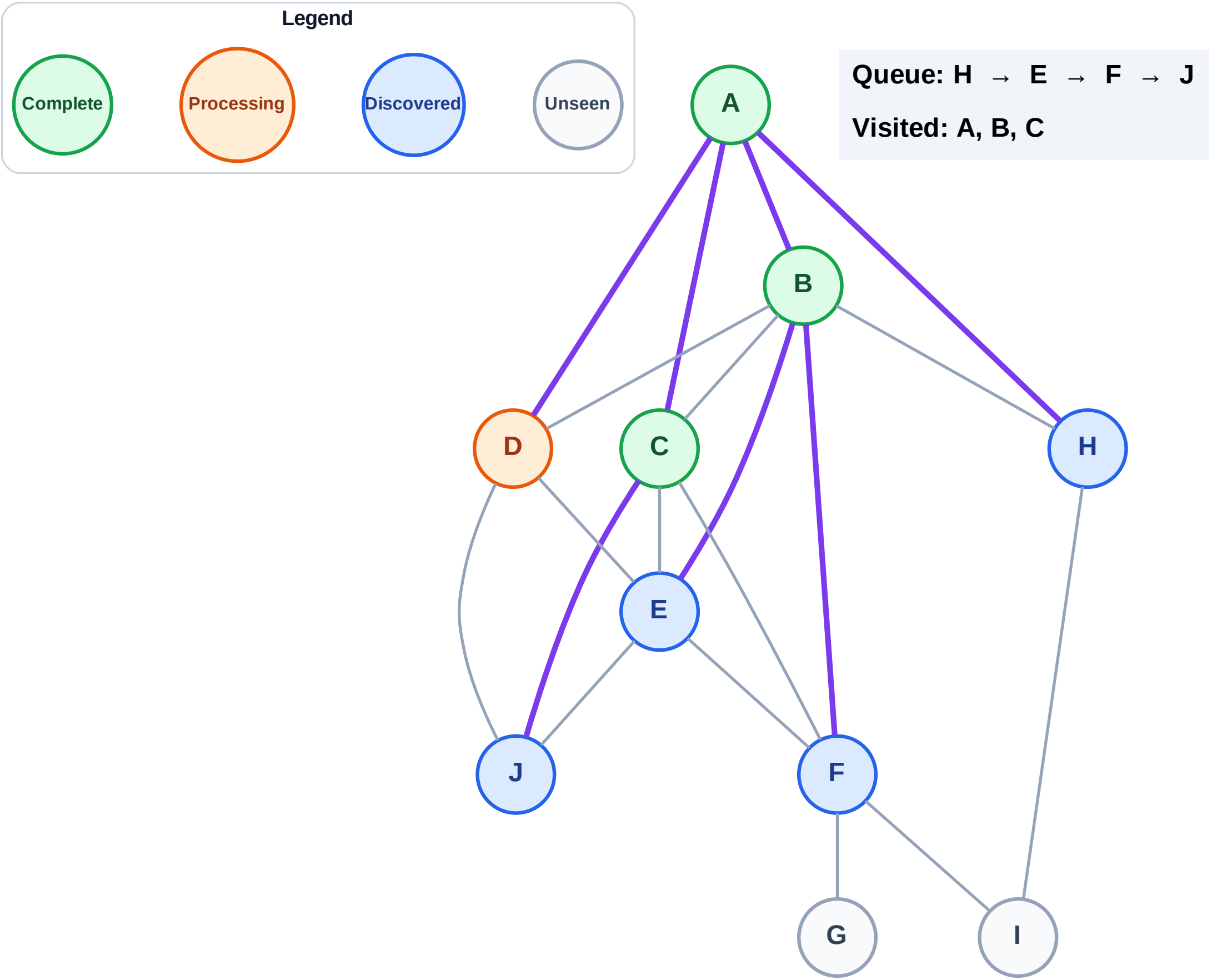
BFS Traversal

Step 7: Discover J from C



BFS Traversal

Step 8: Process node D

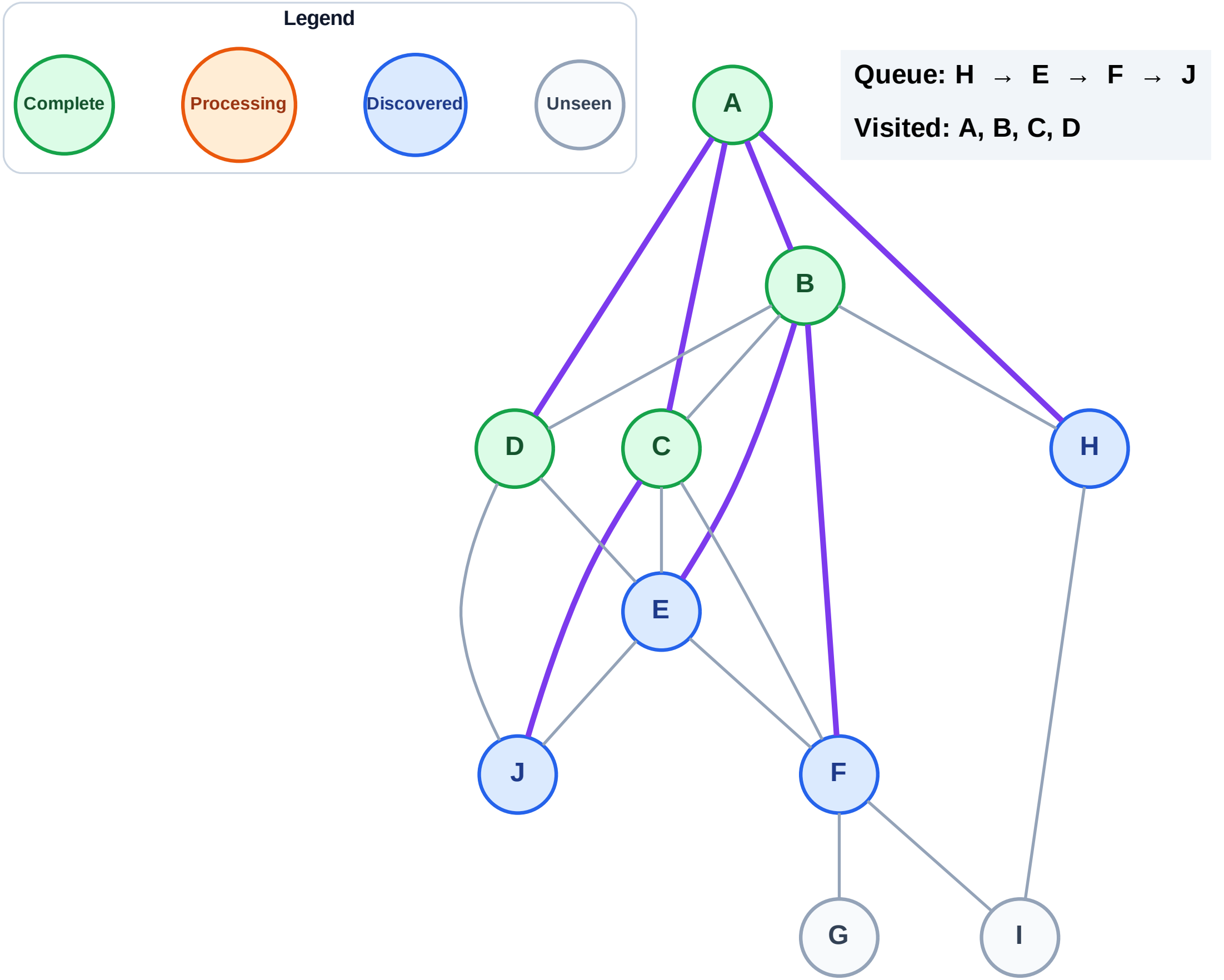


Queue: H → E → F → J

Visited: A, B, C

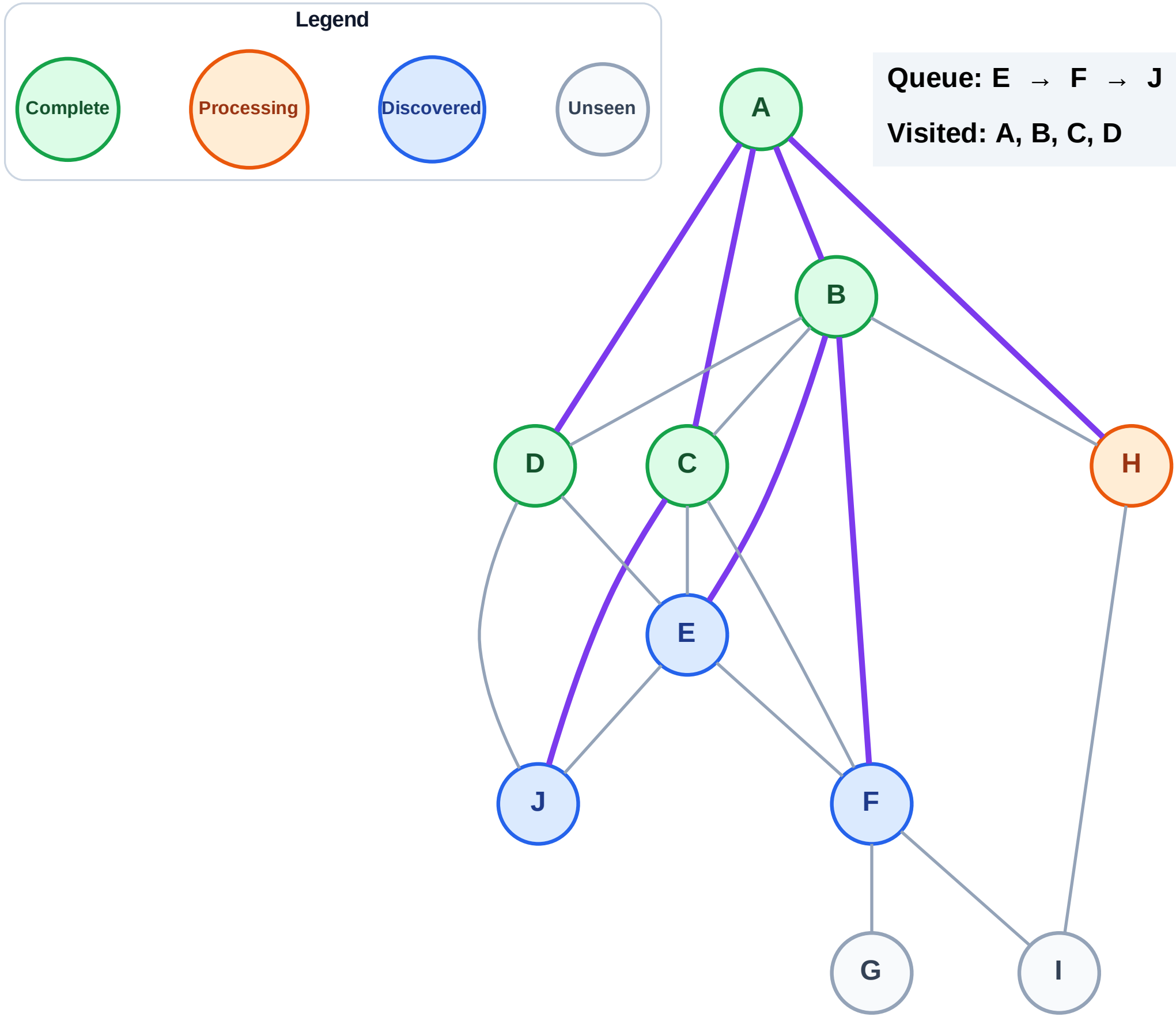
BFS Traversal

Step 9: No new neighbors from D



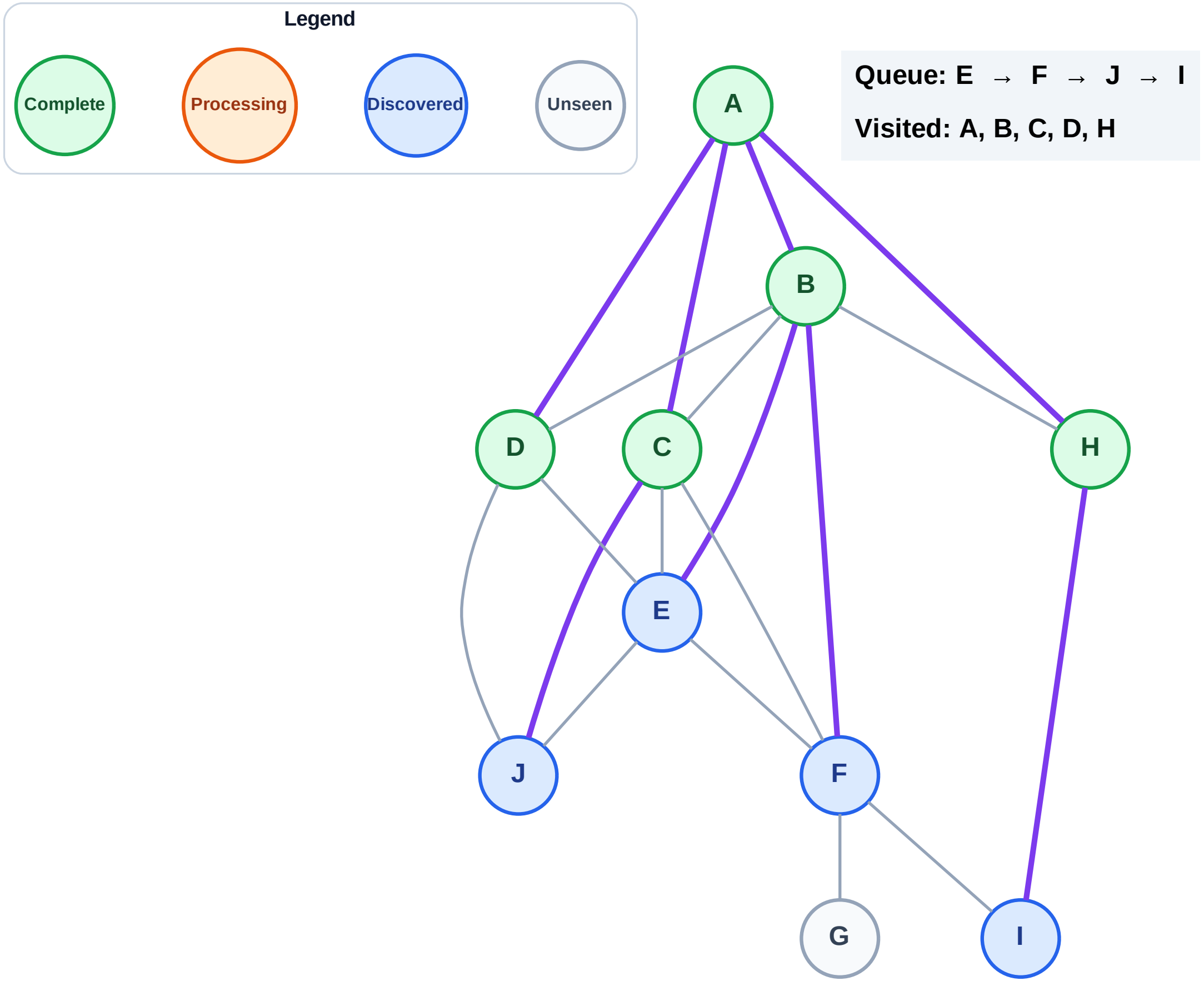
BFS Traversal

Step 10: Process node H



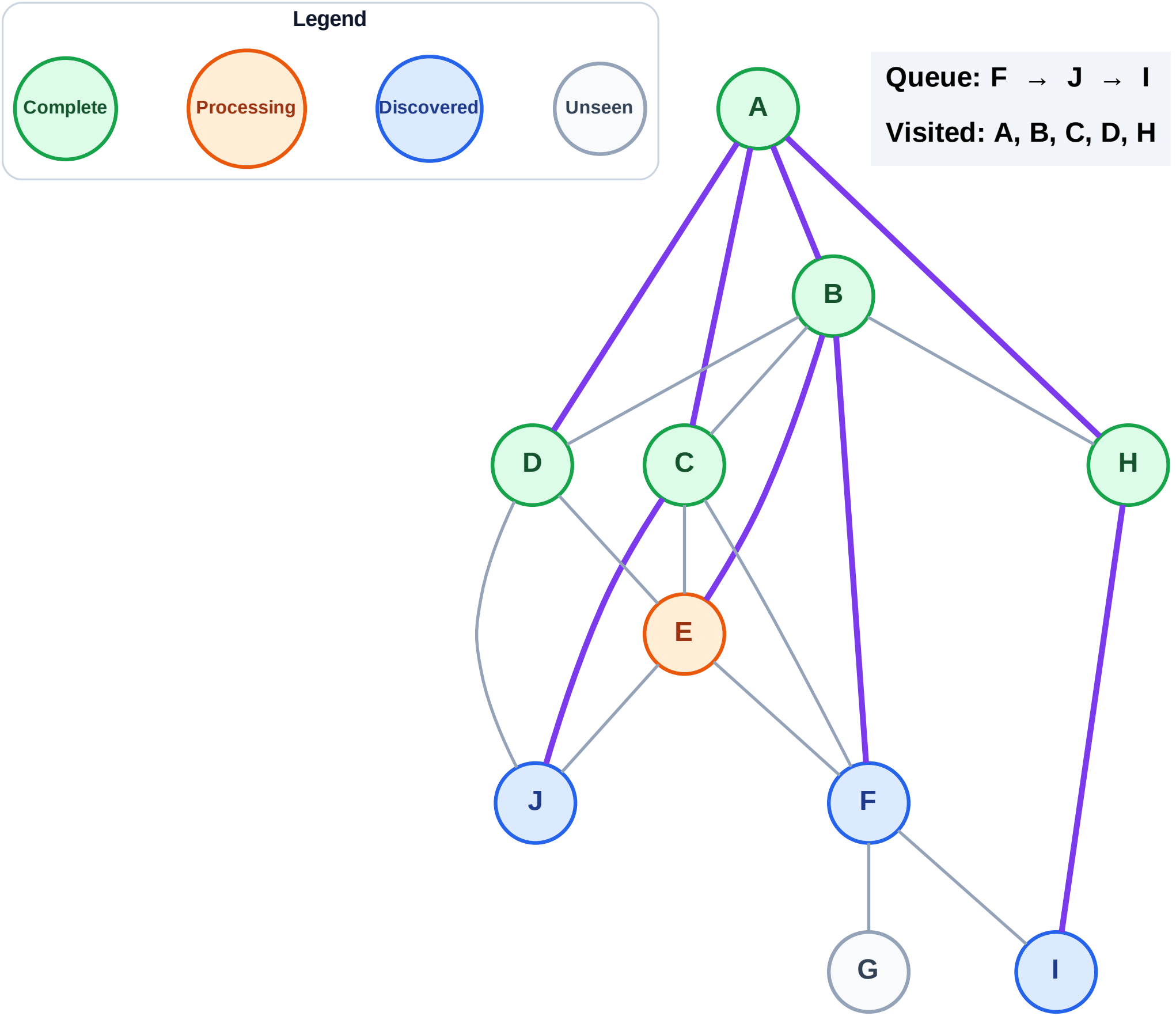
BFS Traversal

Step 11: Discover I from H



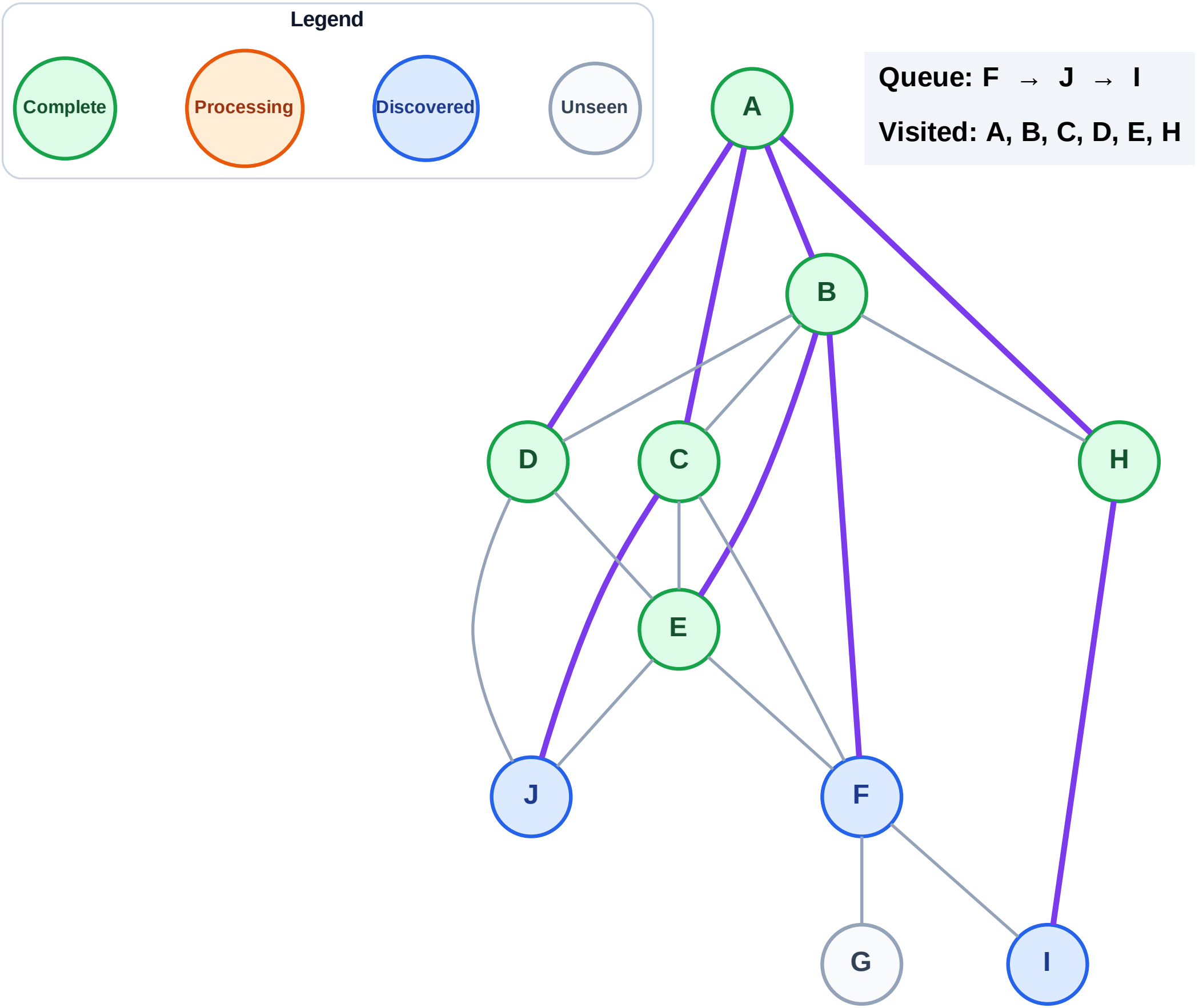
BFS Traversal

Step 12: Process node E



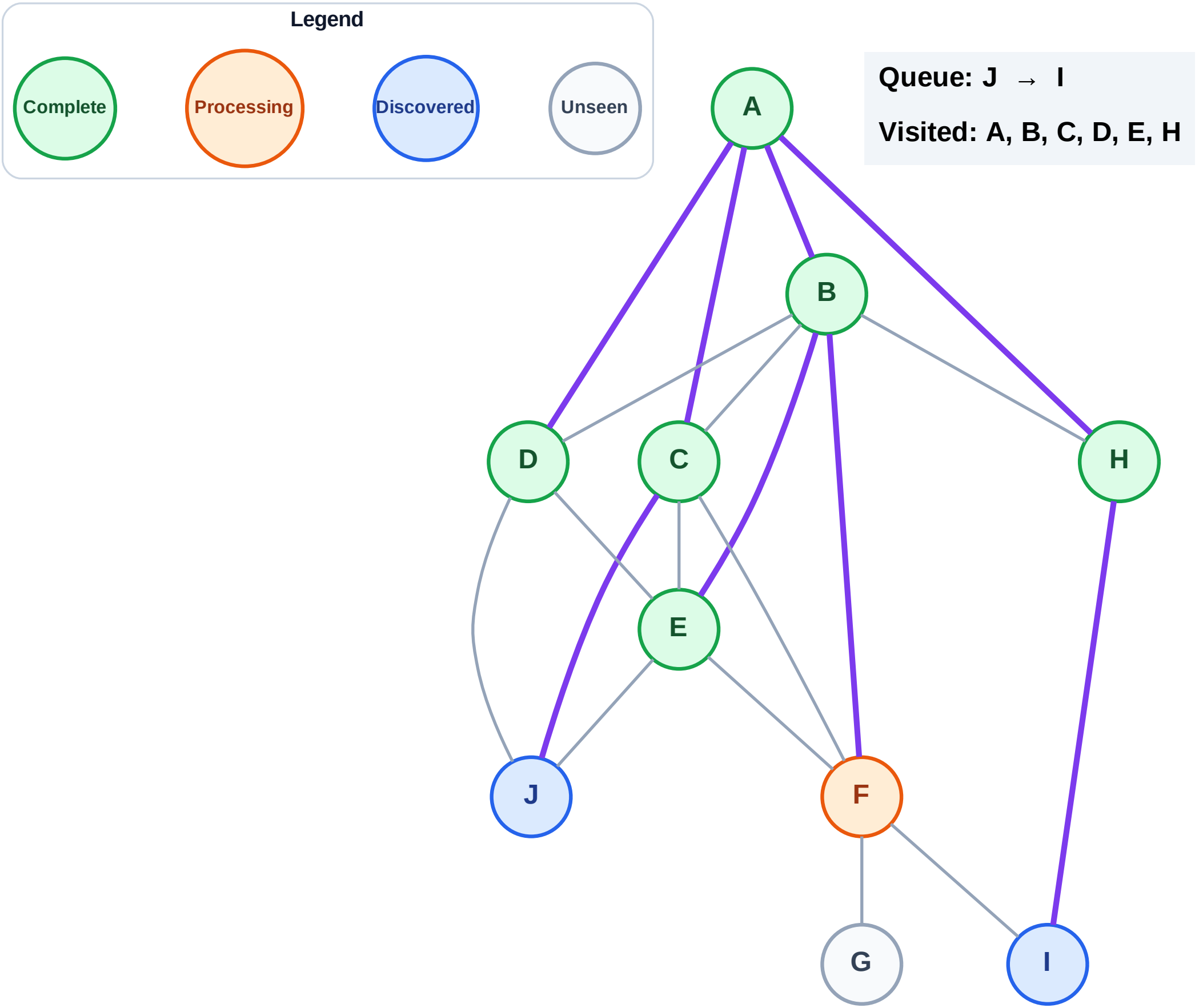
BFS Traversal

Step 13: No new neighbors from E



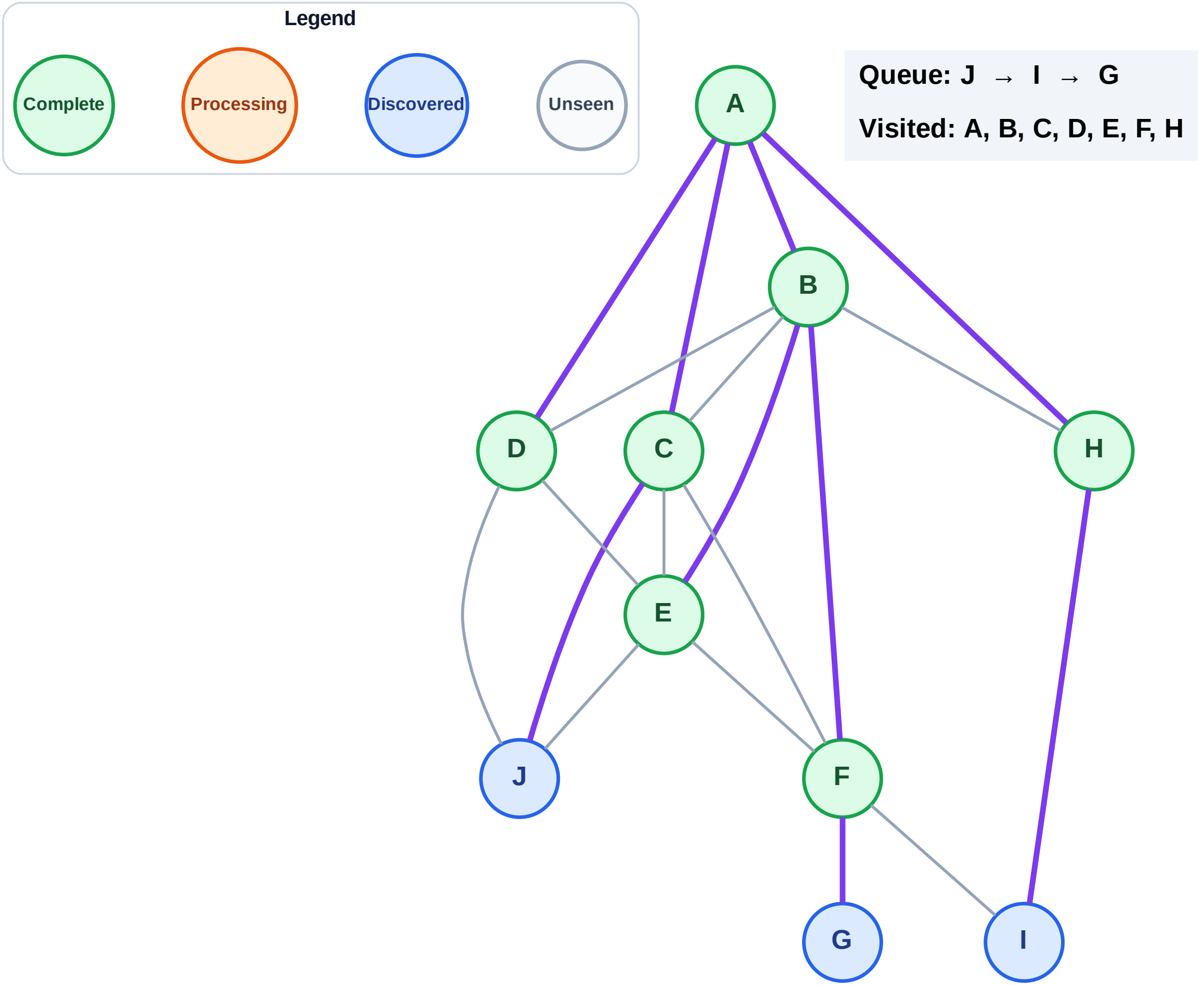
BFS Traversal

Step 14: Process node F



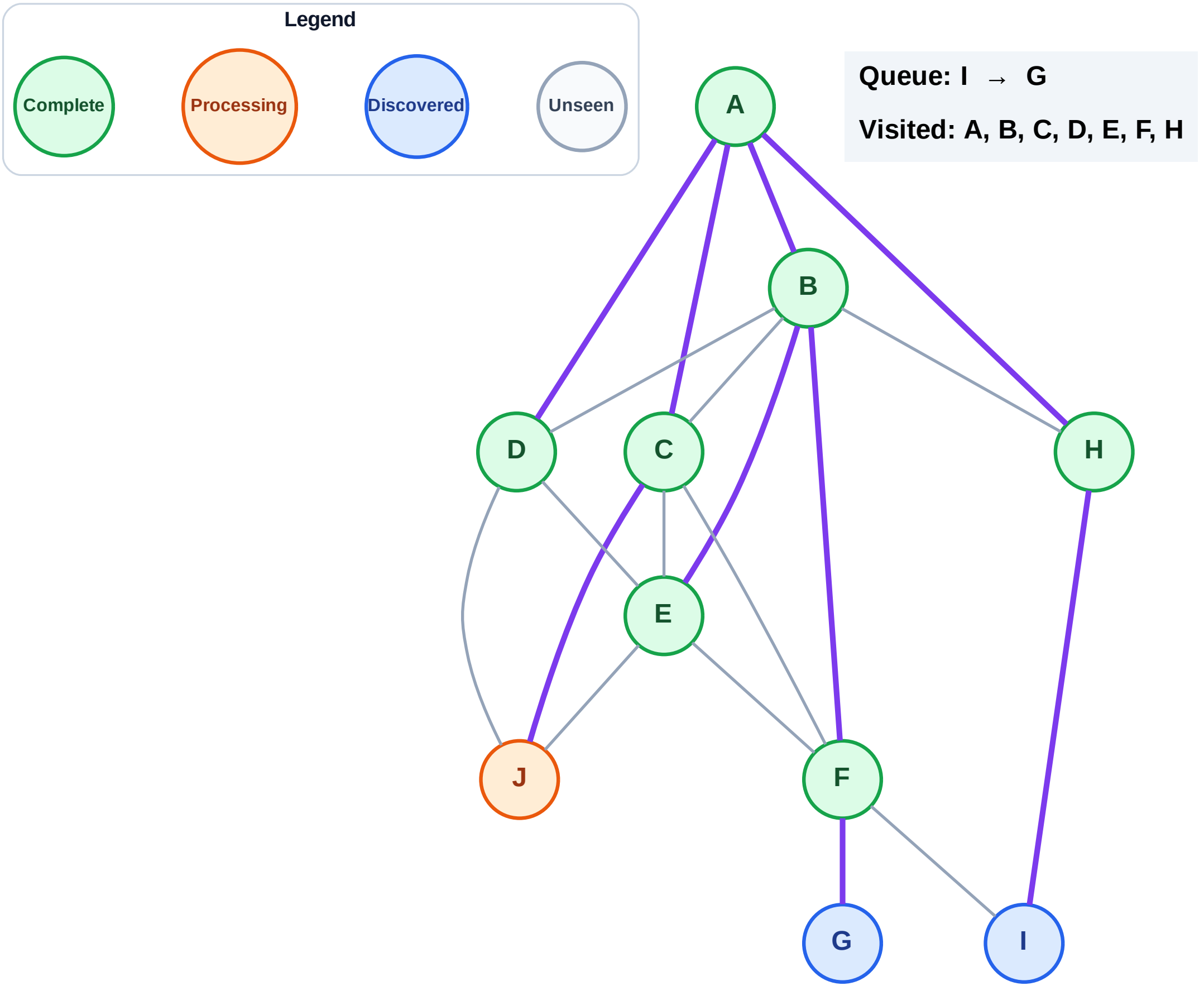
BFS Traversal

Step 15: Discover G from F



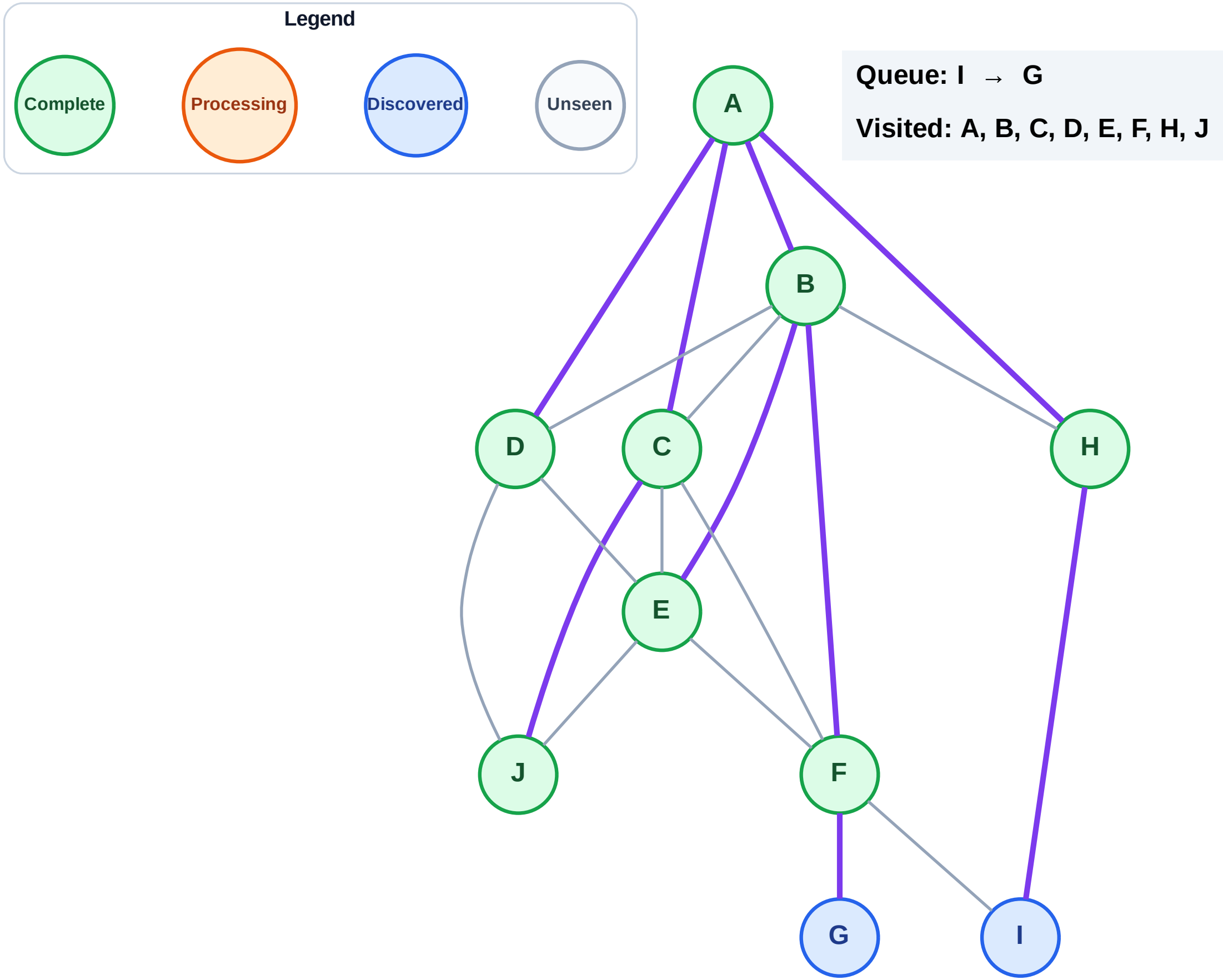
BFS Traversal

Step 16: Process node J



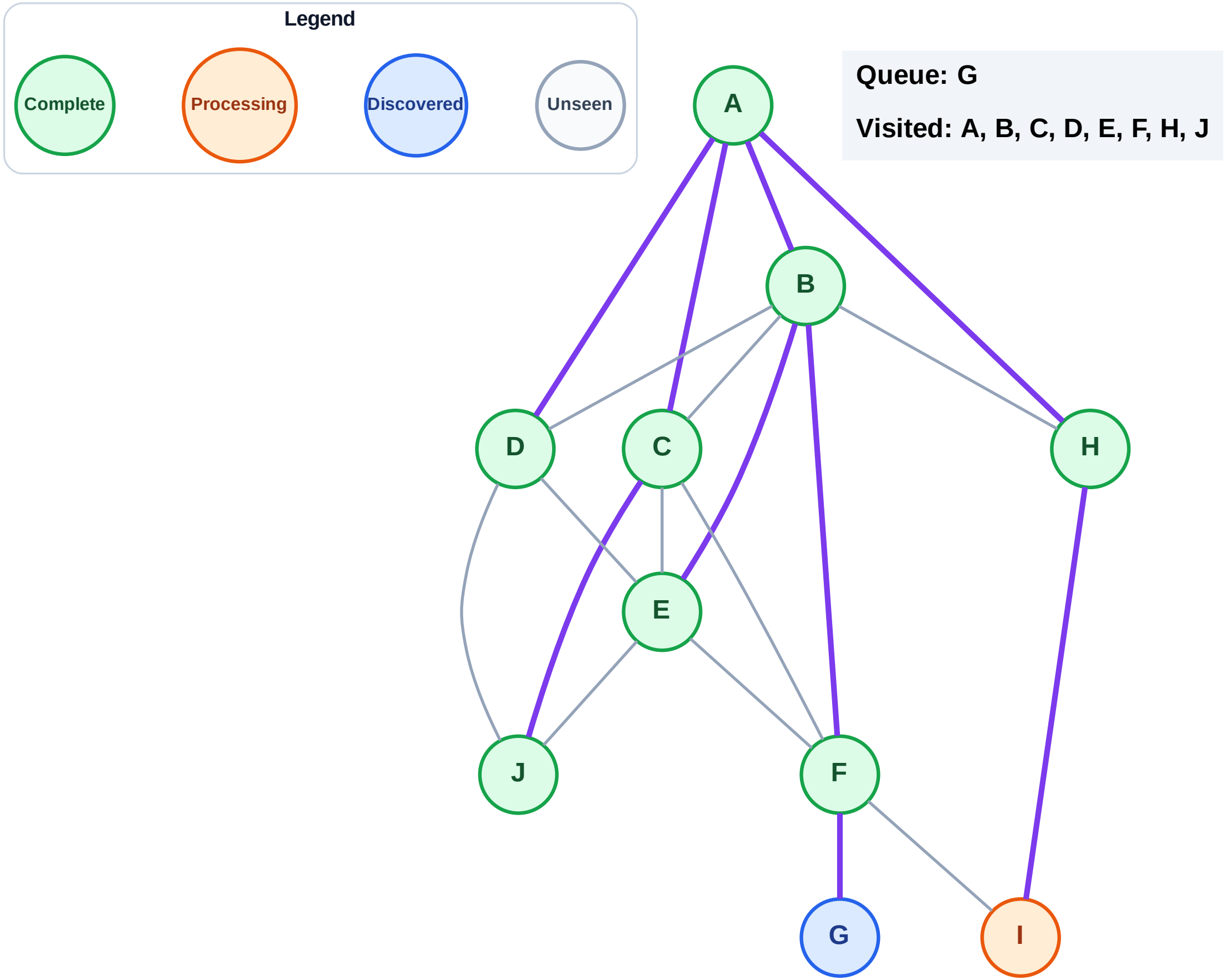
BFS Traversal

Step 17: No new neighbors from J



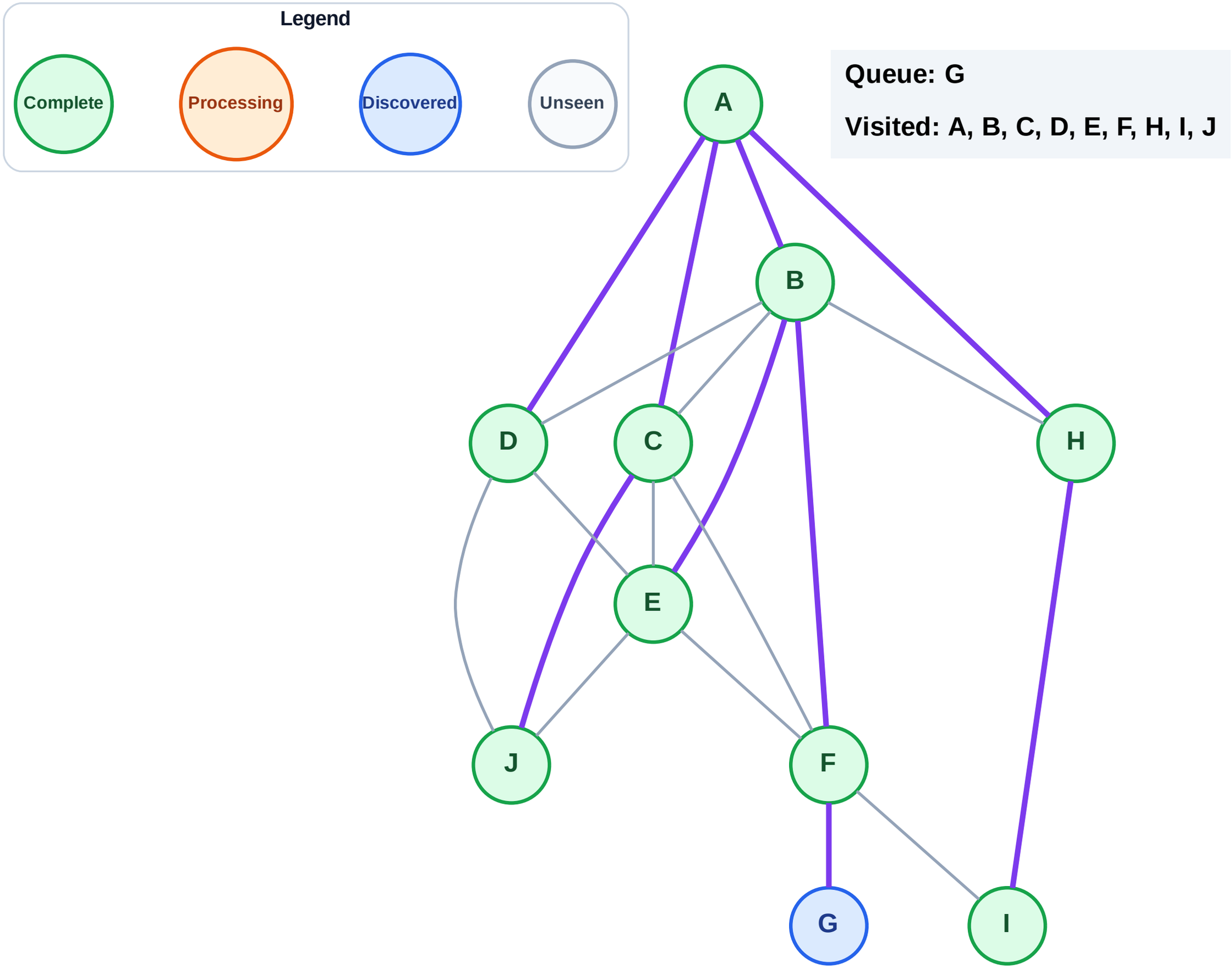
BFS Traversal

Step 18: Process node I



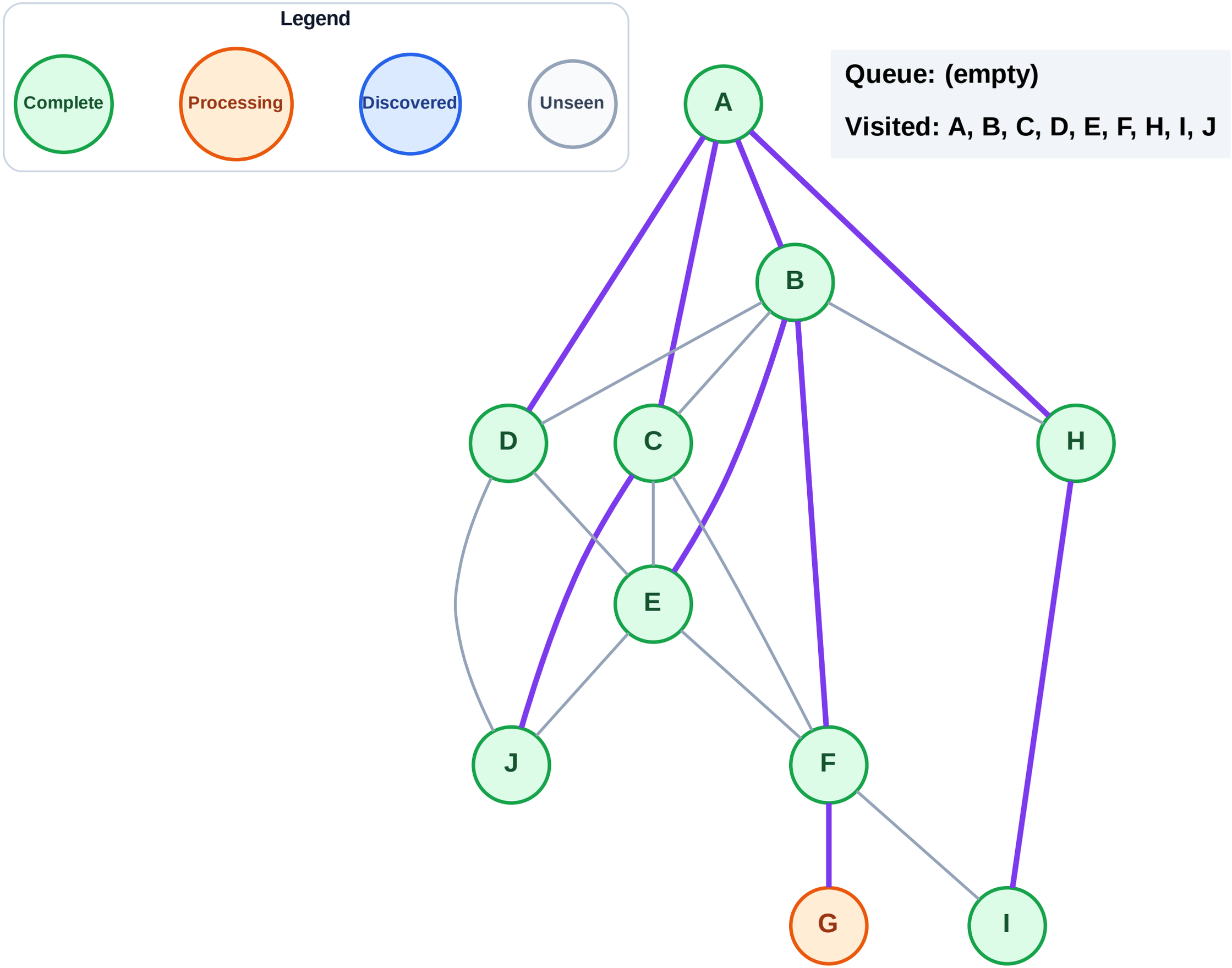
BFS Traversal

Step 19: No new neighbors from I



BFS Traversal

Step 20: Process node G

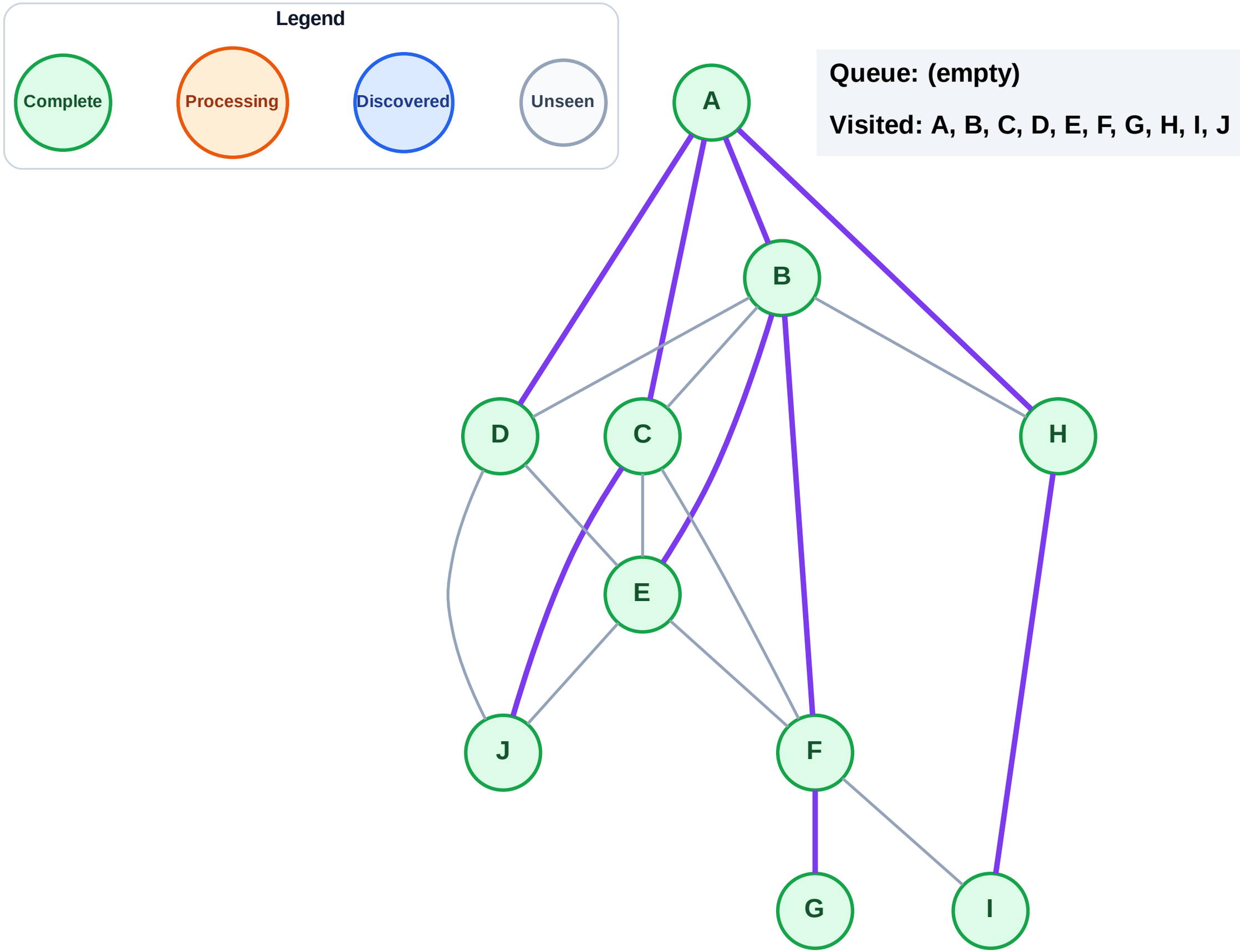


Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J

BFS Traversal

Step 21: No new neighbors from G

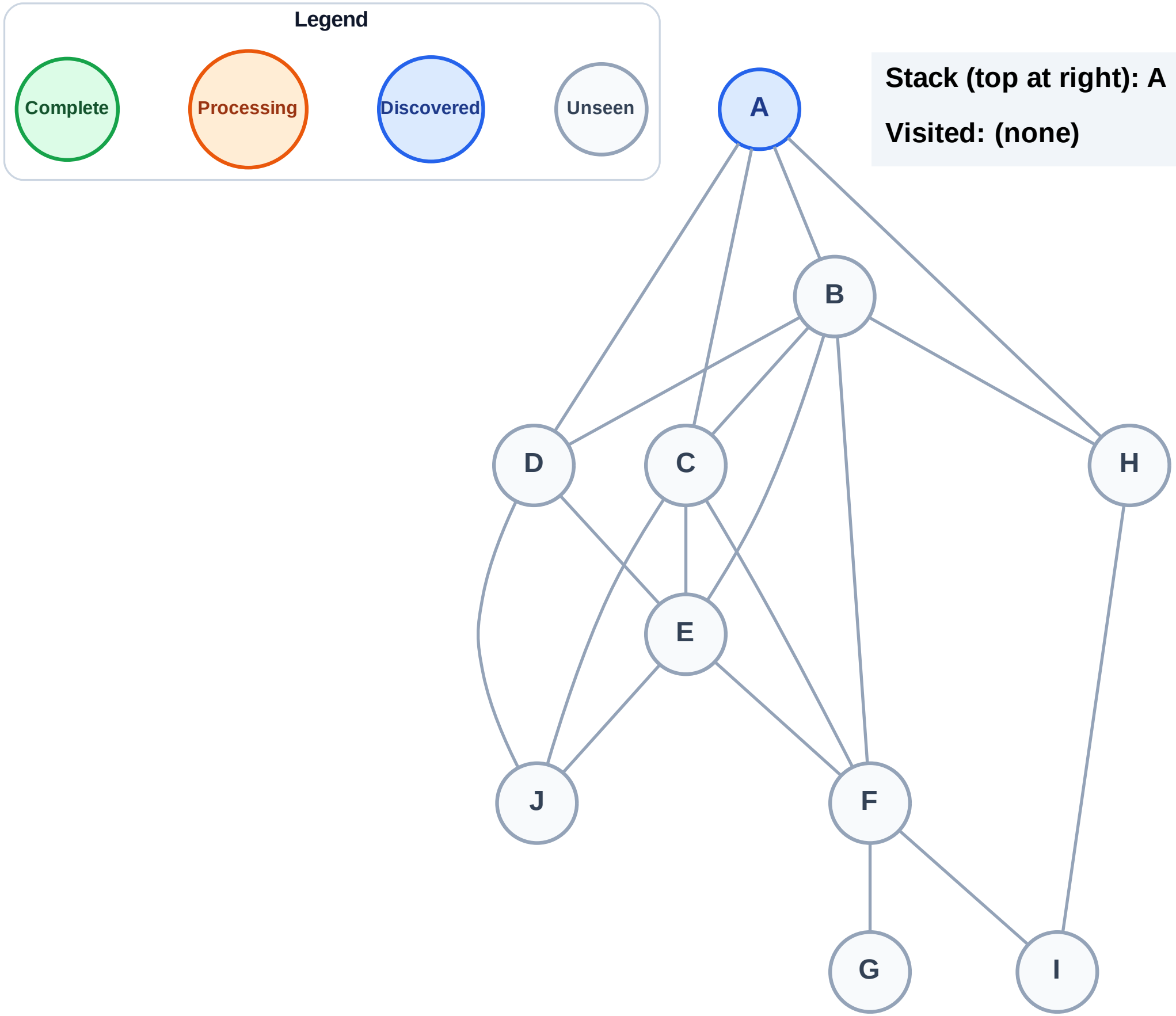


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

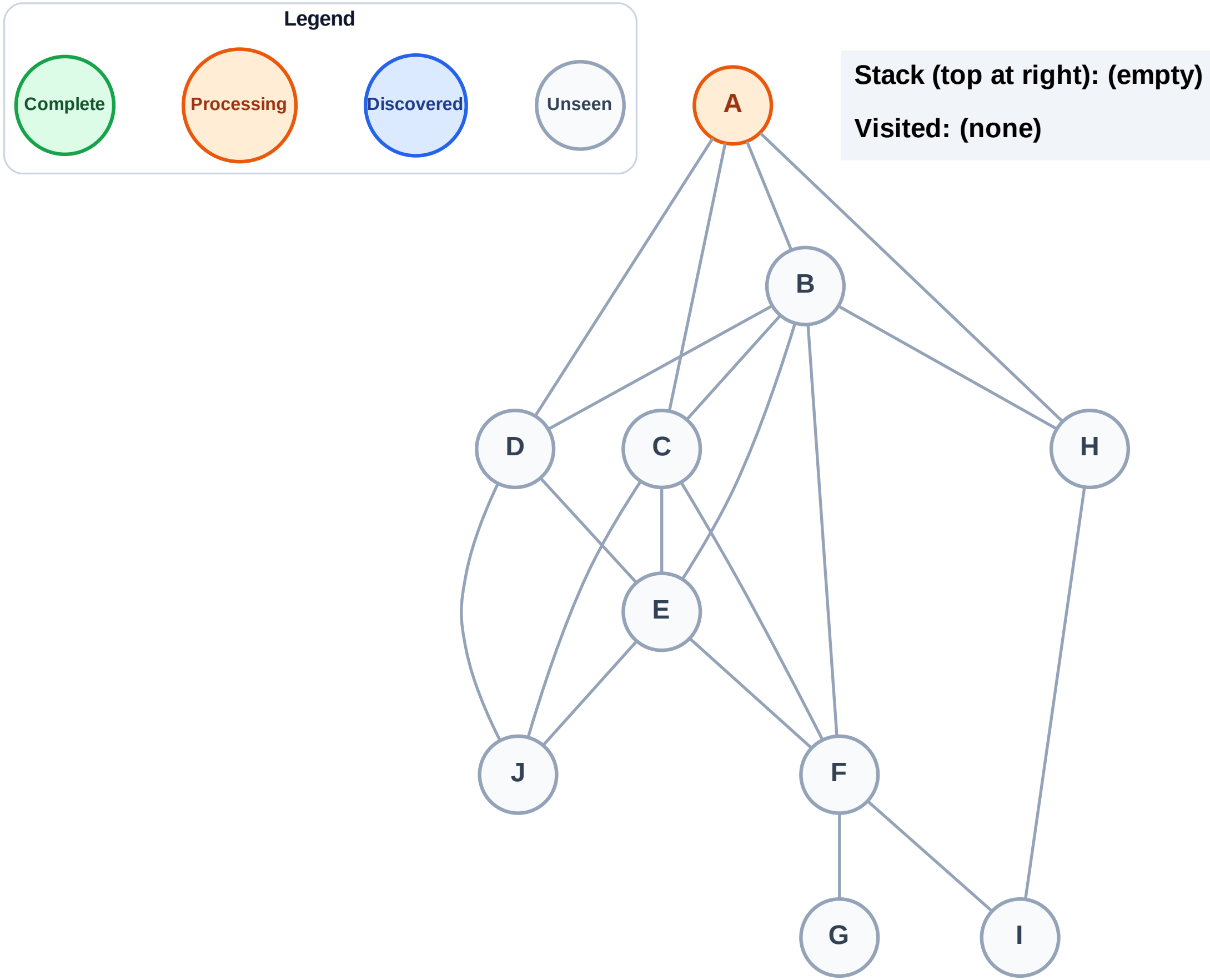
DFS Traversal

Step 1: Start at A



DFS Traversal

Step 2: Process node A

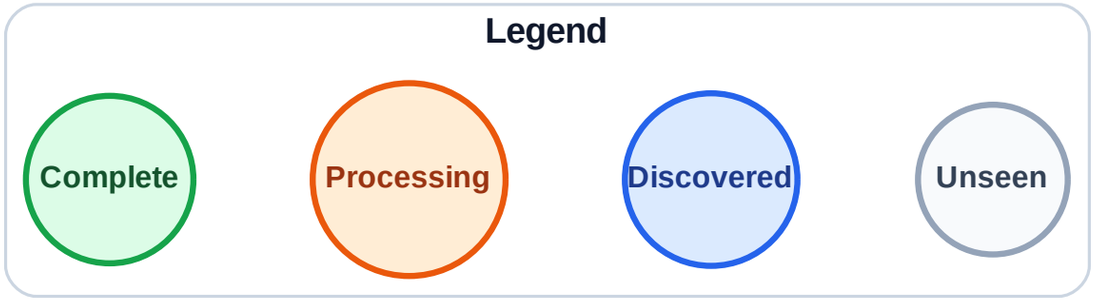


Stack (top at right): (empty)

Visited: (none)

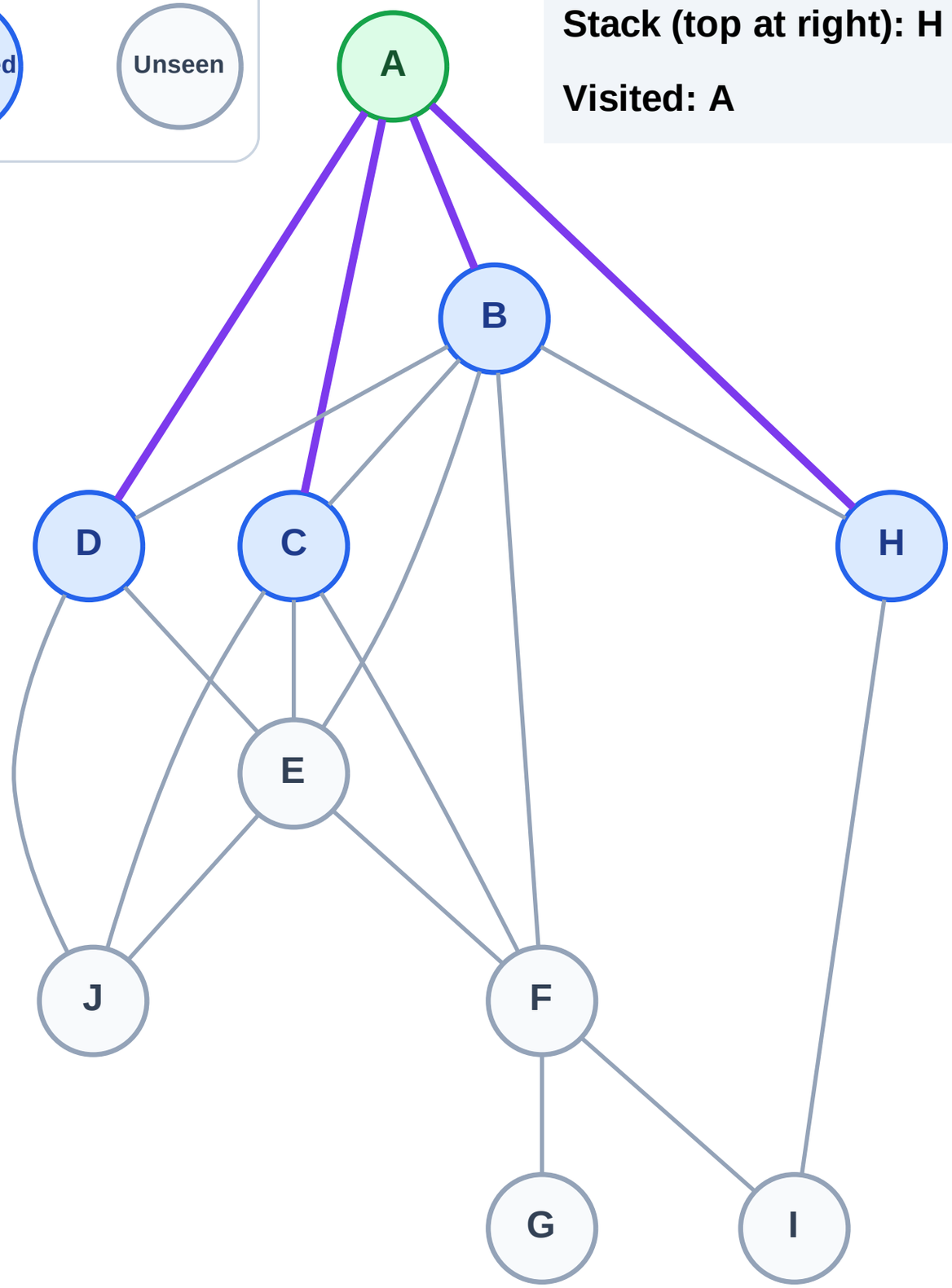
DFS Traversal

Step 3: Discover H, D, C, B from A



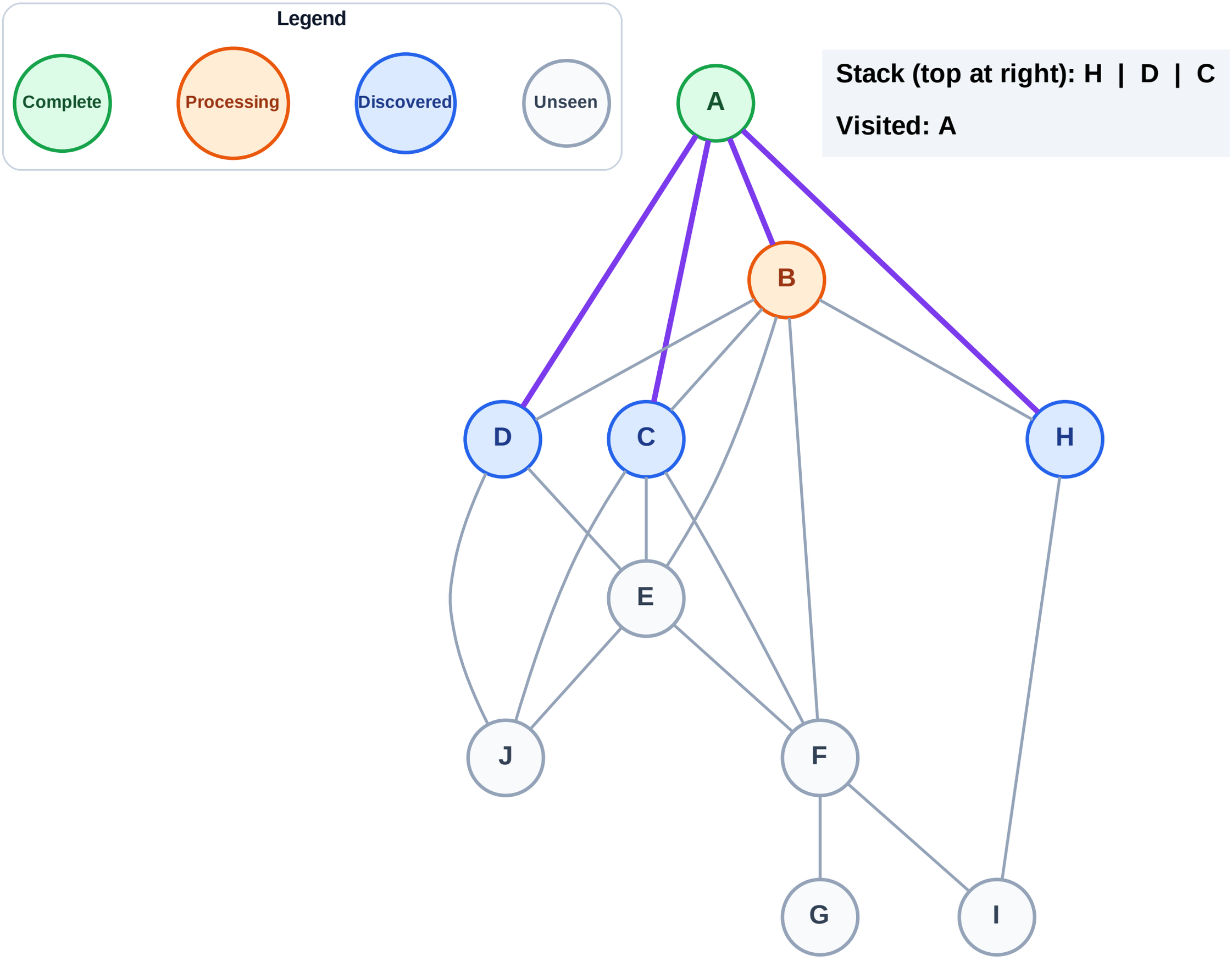
Stack (top at right): H | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

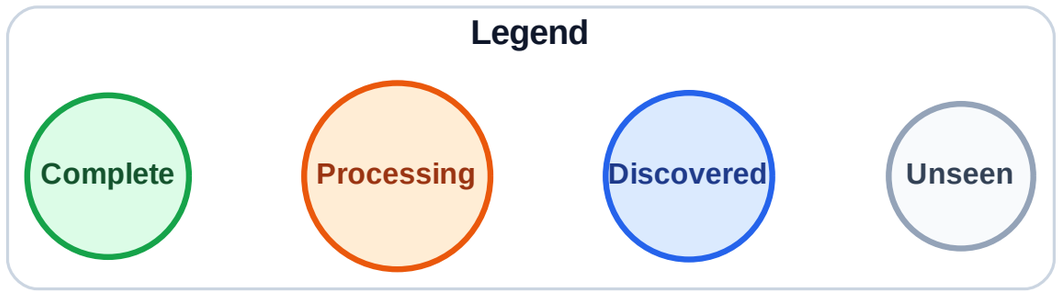


Stack (top at right): H | D | C

Visited: A

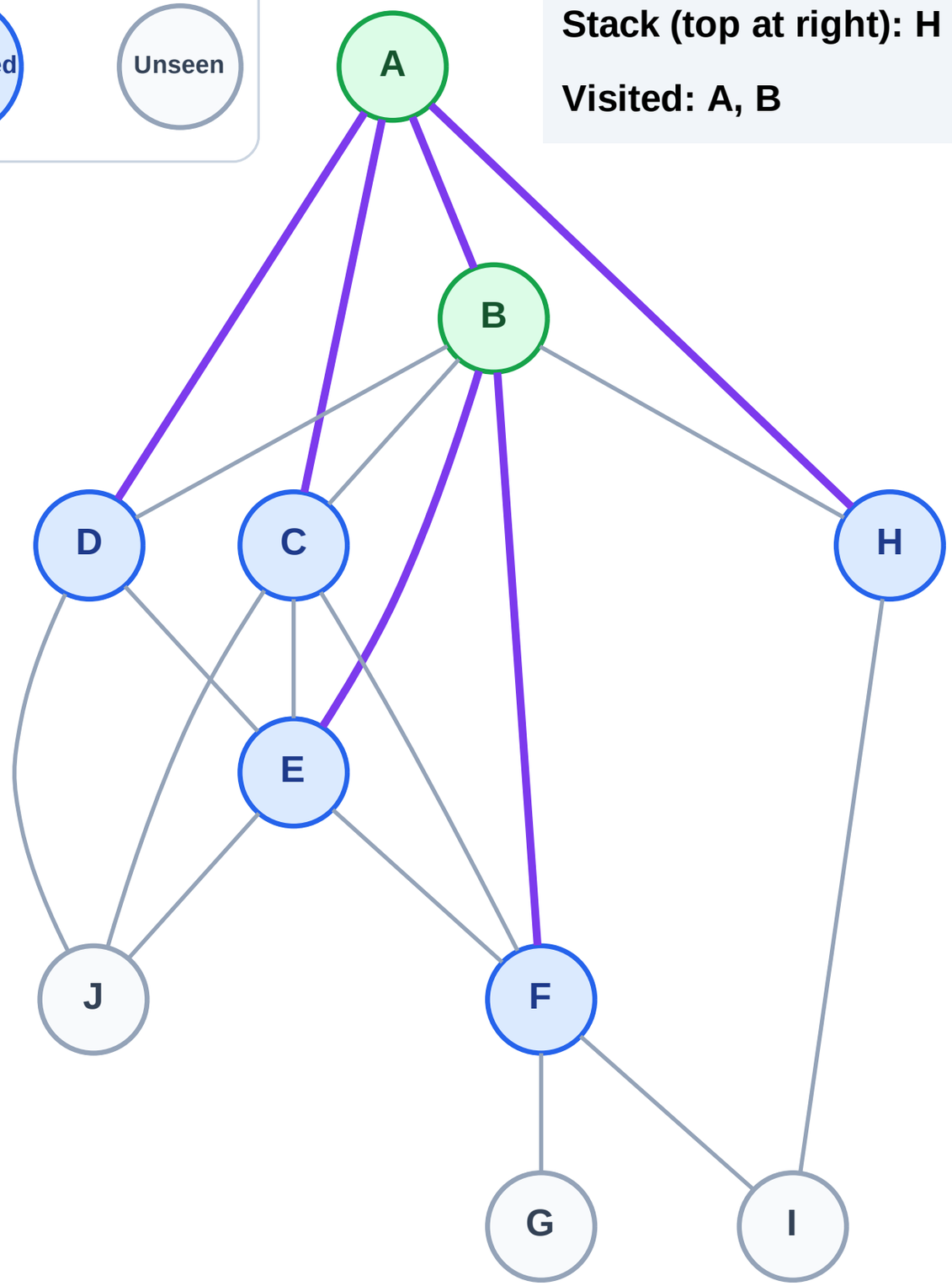
DFS Traversal

Step 5: Discover F, E from B



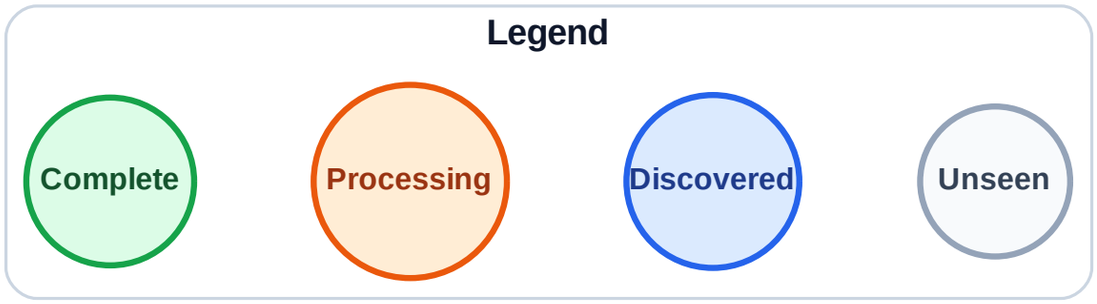
Stack (top at right): H | D | C | F | E

Visited: A, B



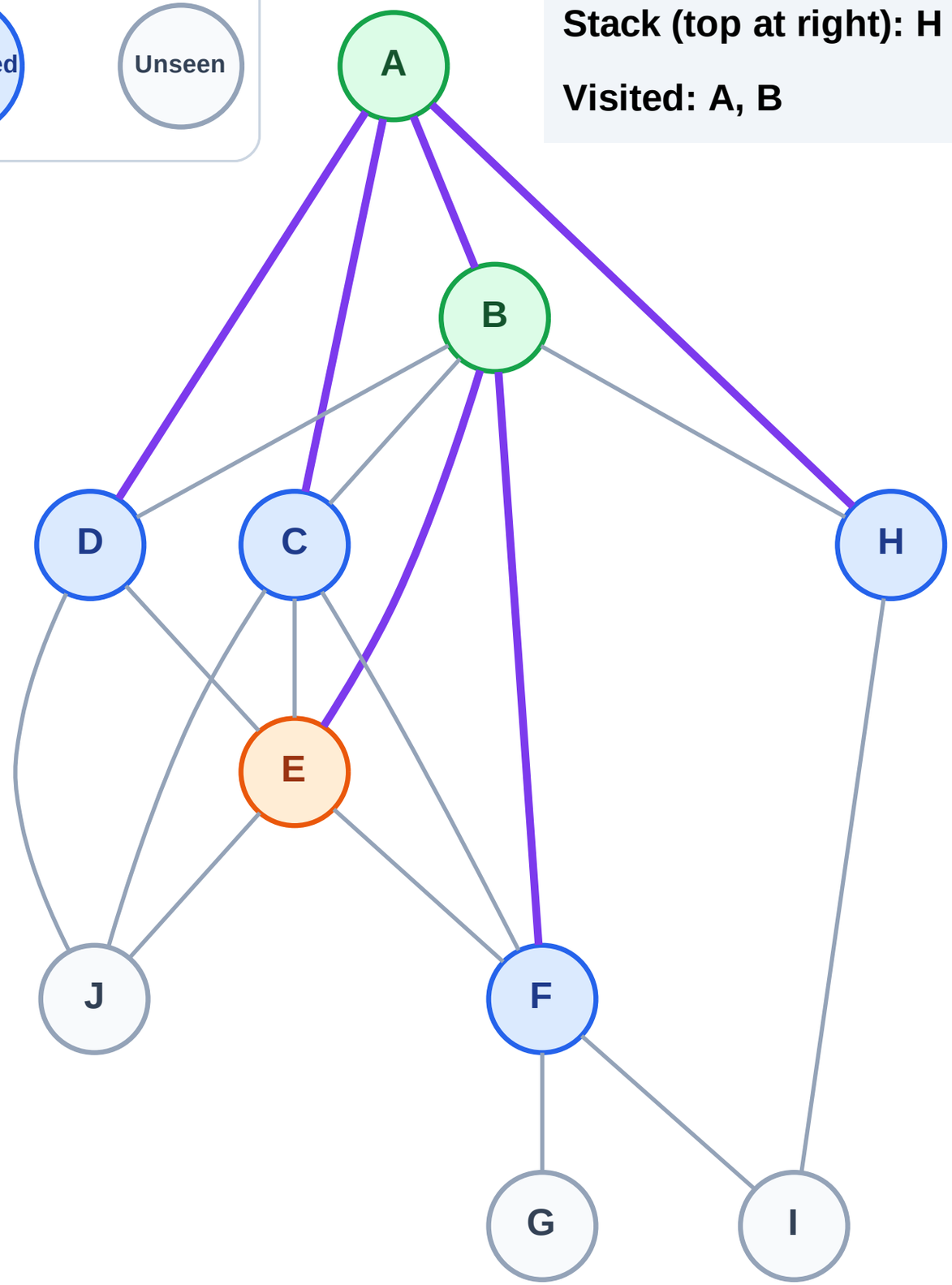
DFS Traversal

Step 6: Process node E



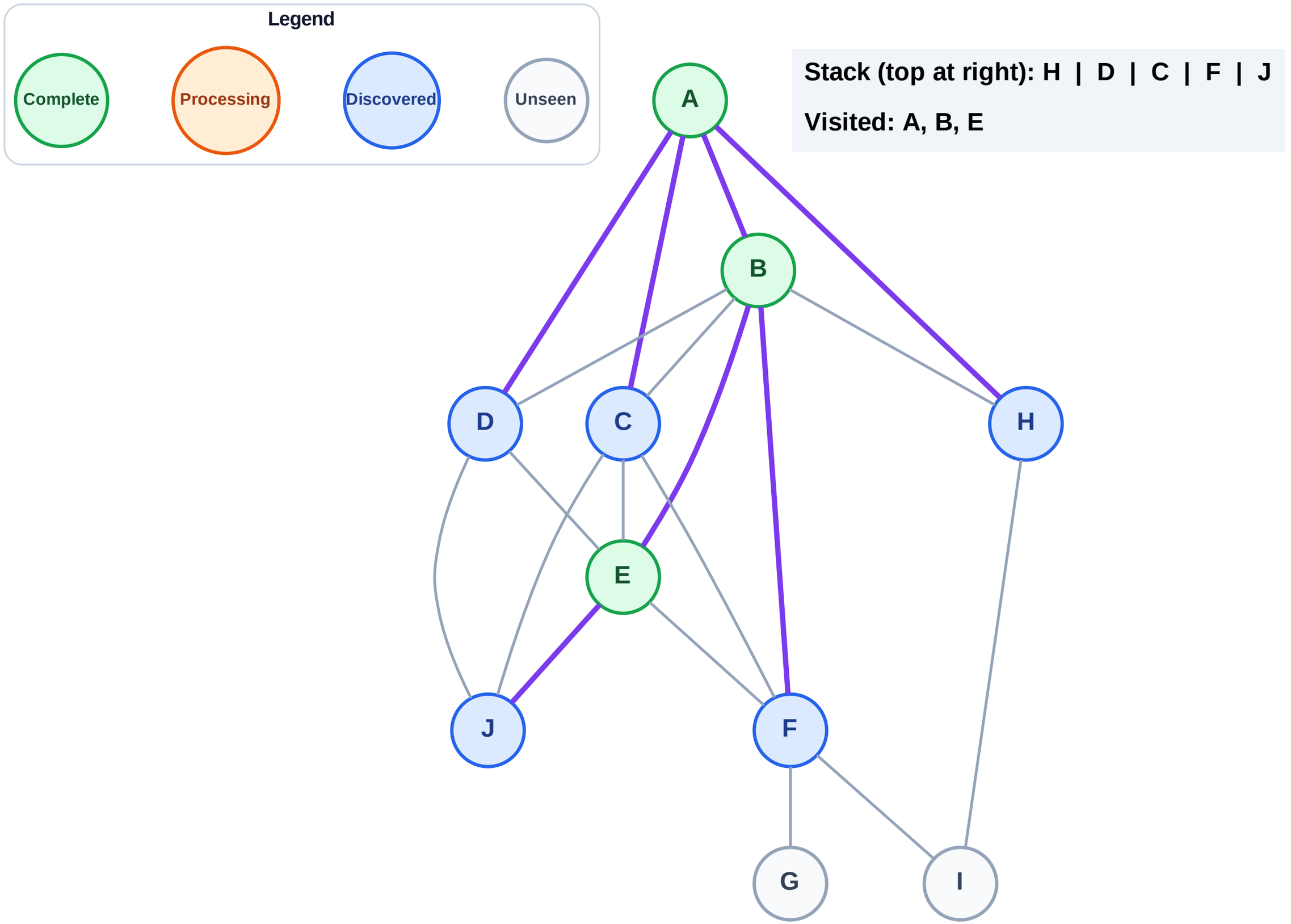
Stack (top at right): H | D | C | F

Visited: A, B



DFS Traversal

Step 7: Discover J from E

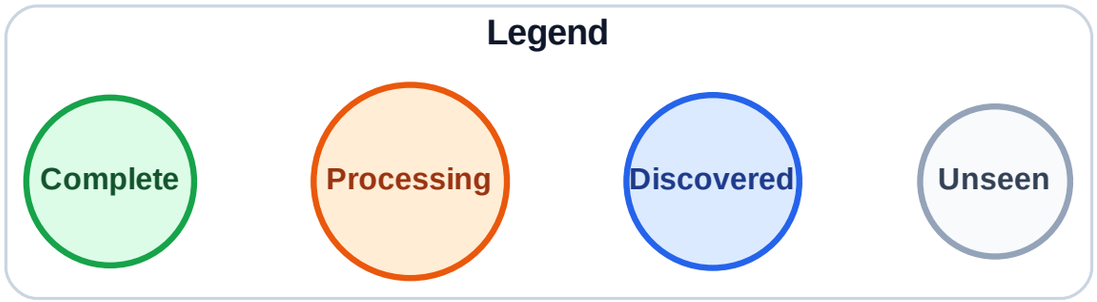


Stack (top at right): H | D | C | F | J

Visited: A, B, E

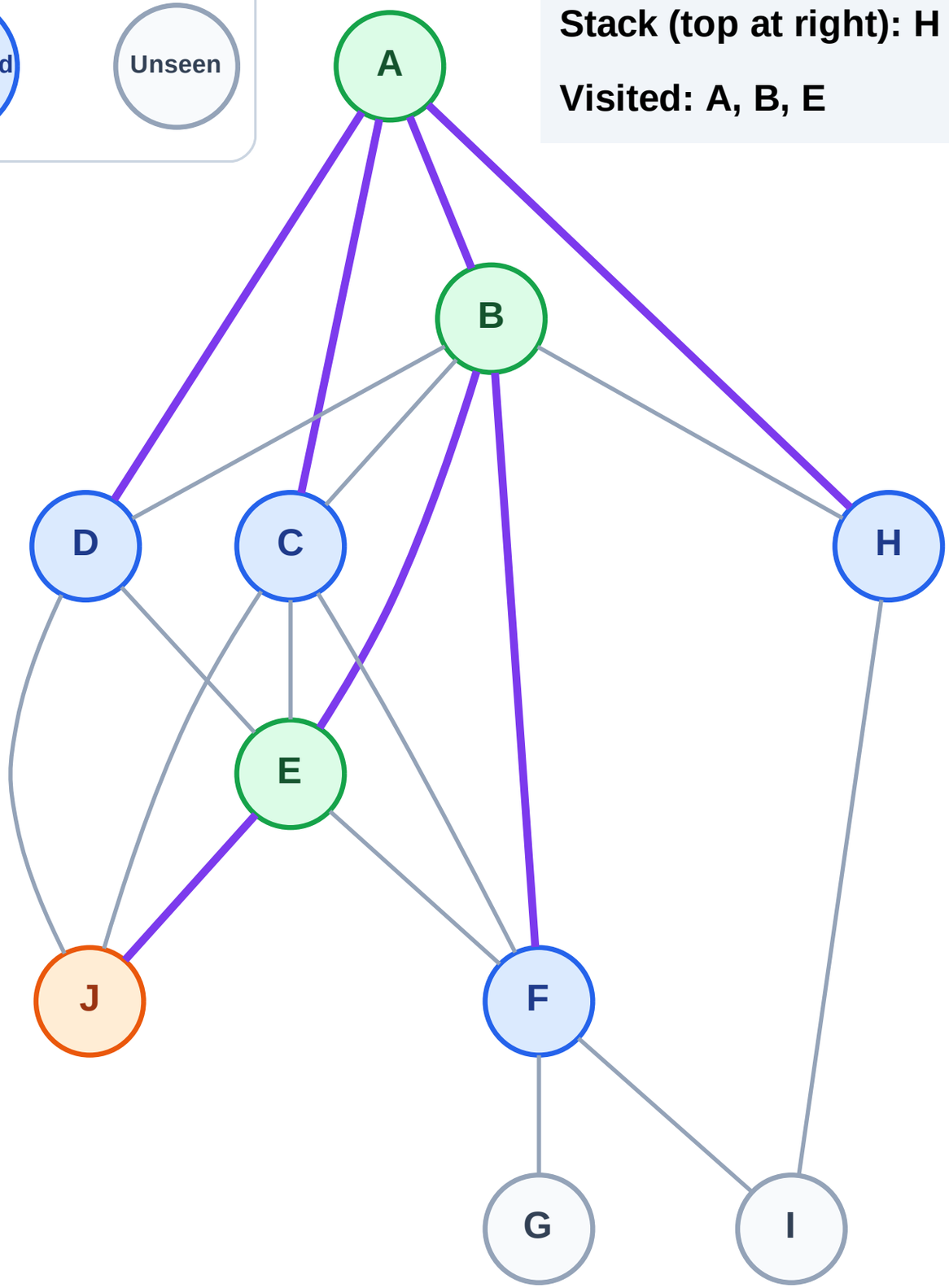
DFS Traversal

Step 8: Process node J



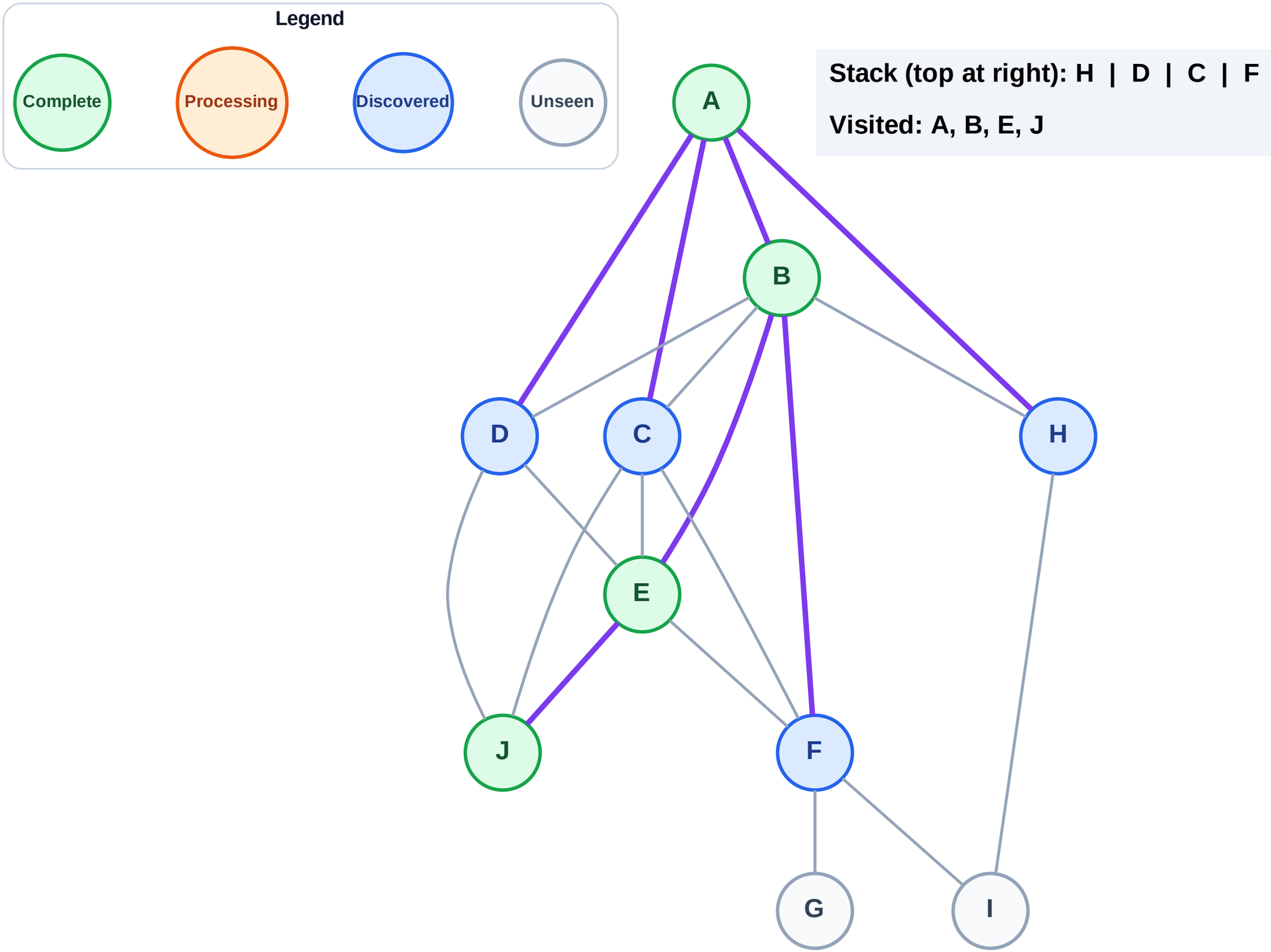
Stack (top at right): H | D | C | F

Visited: A, B, E



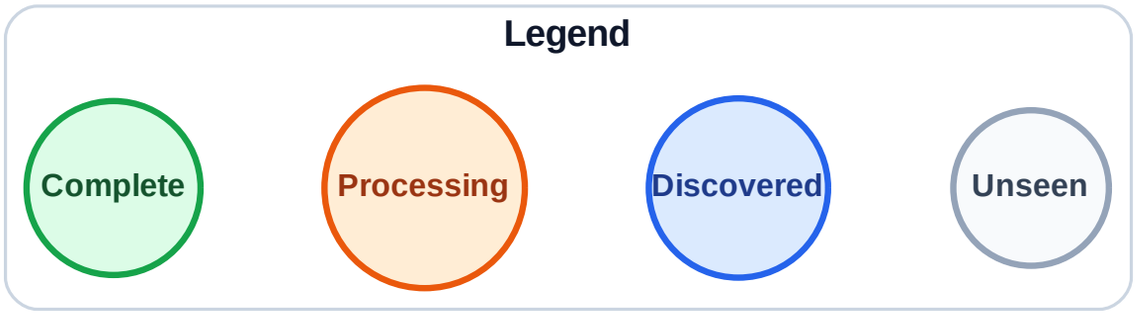
DFS Traversal

Step 9: No new neighbors from J



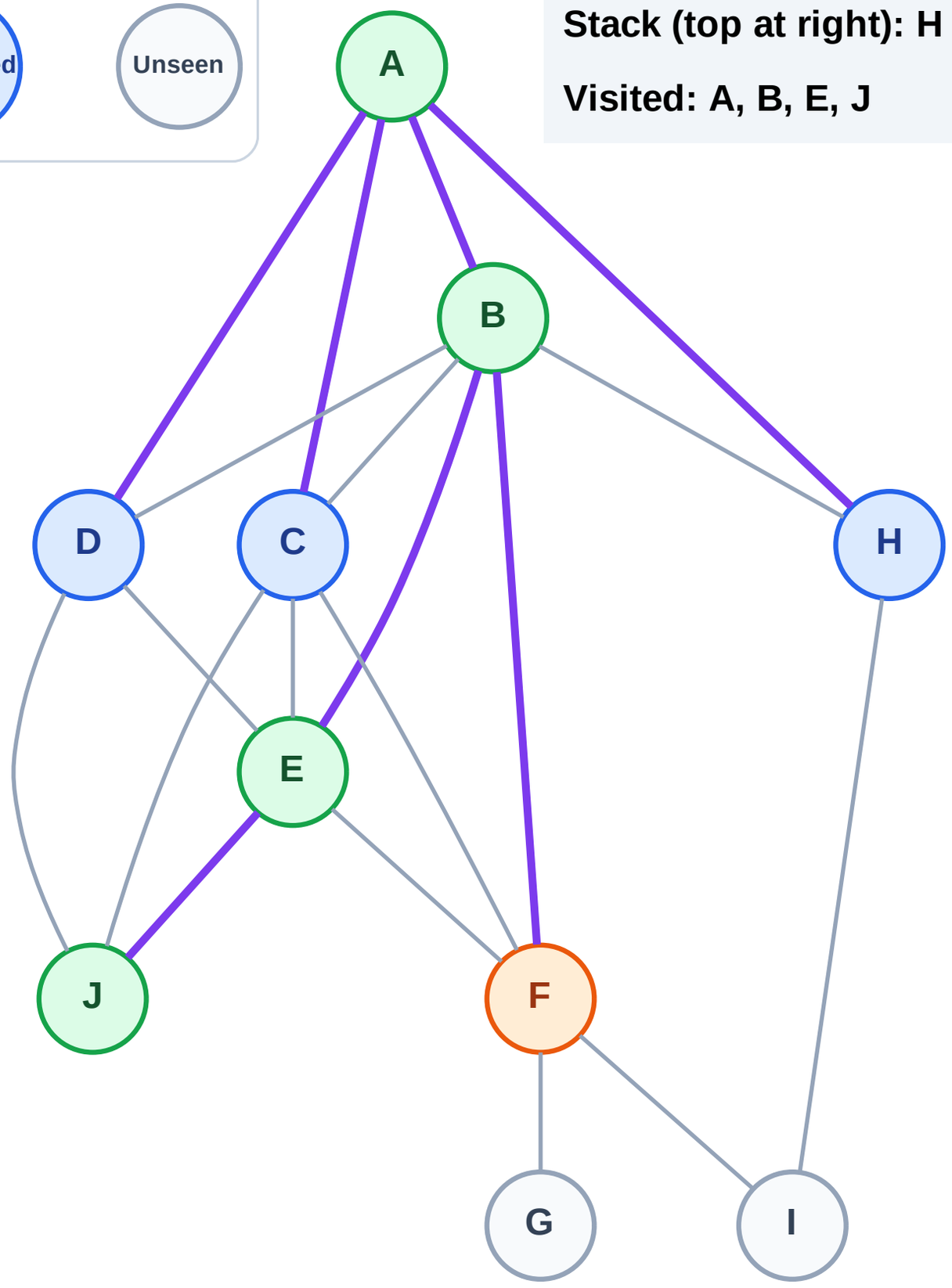
DFS Traversal

Step 10: Process node F



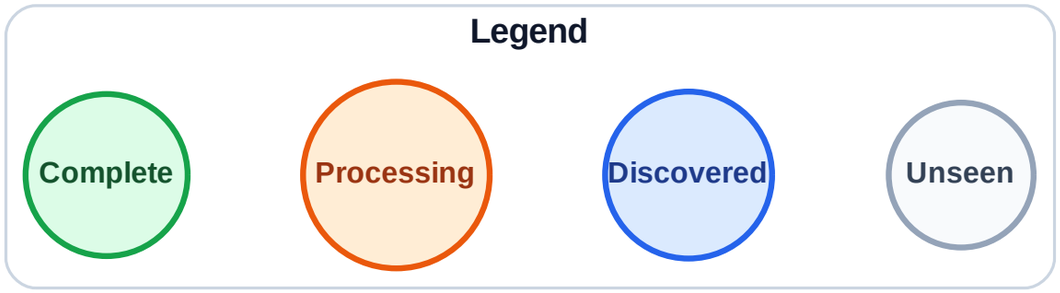
Stack (top at right): H | D | C

Visited: A, B, E, J



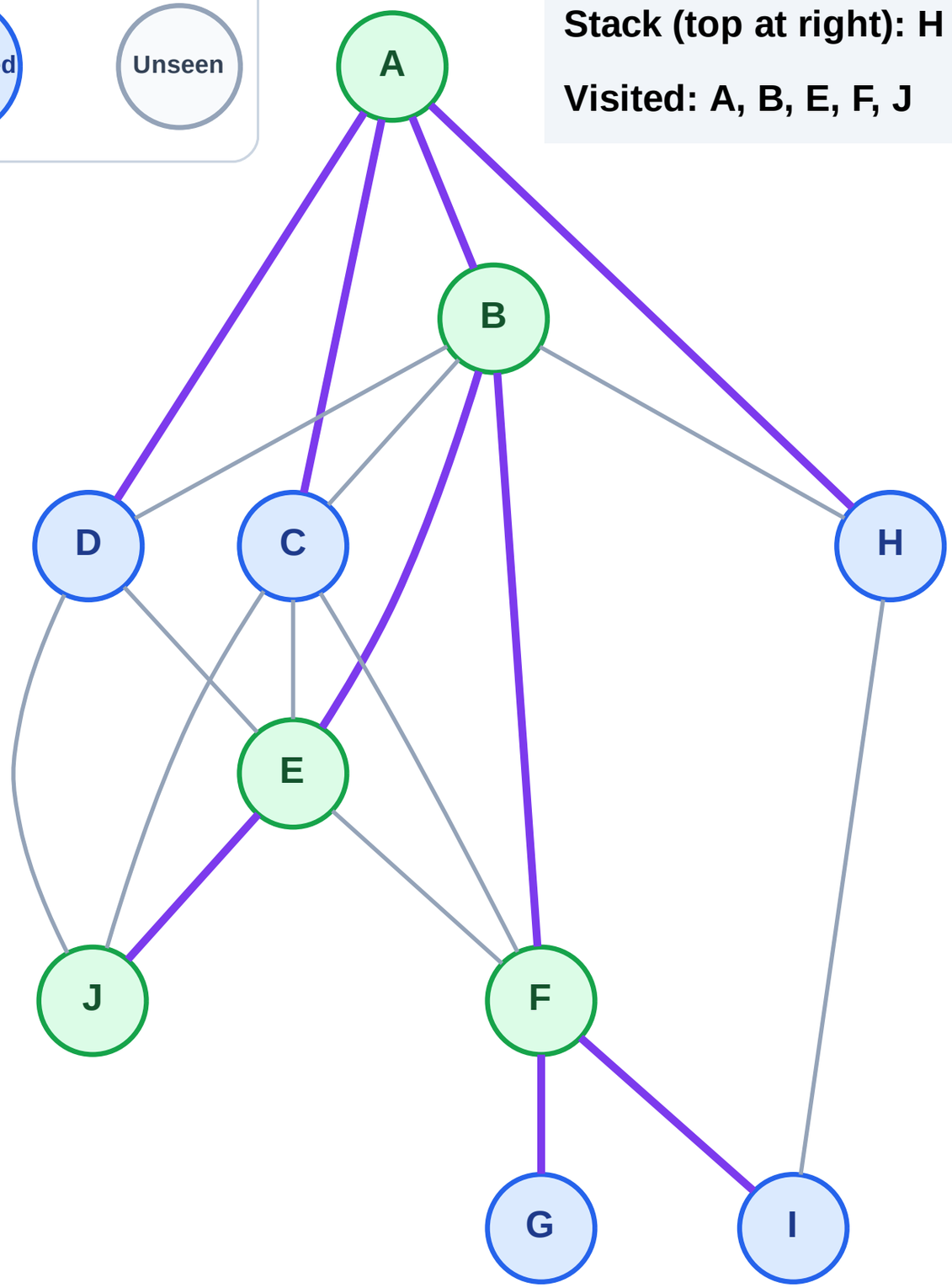
DFS Traversal

Step 11: Discover I, G from F



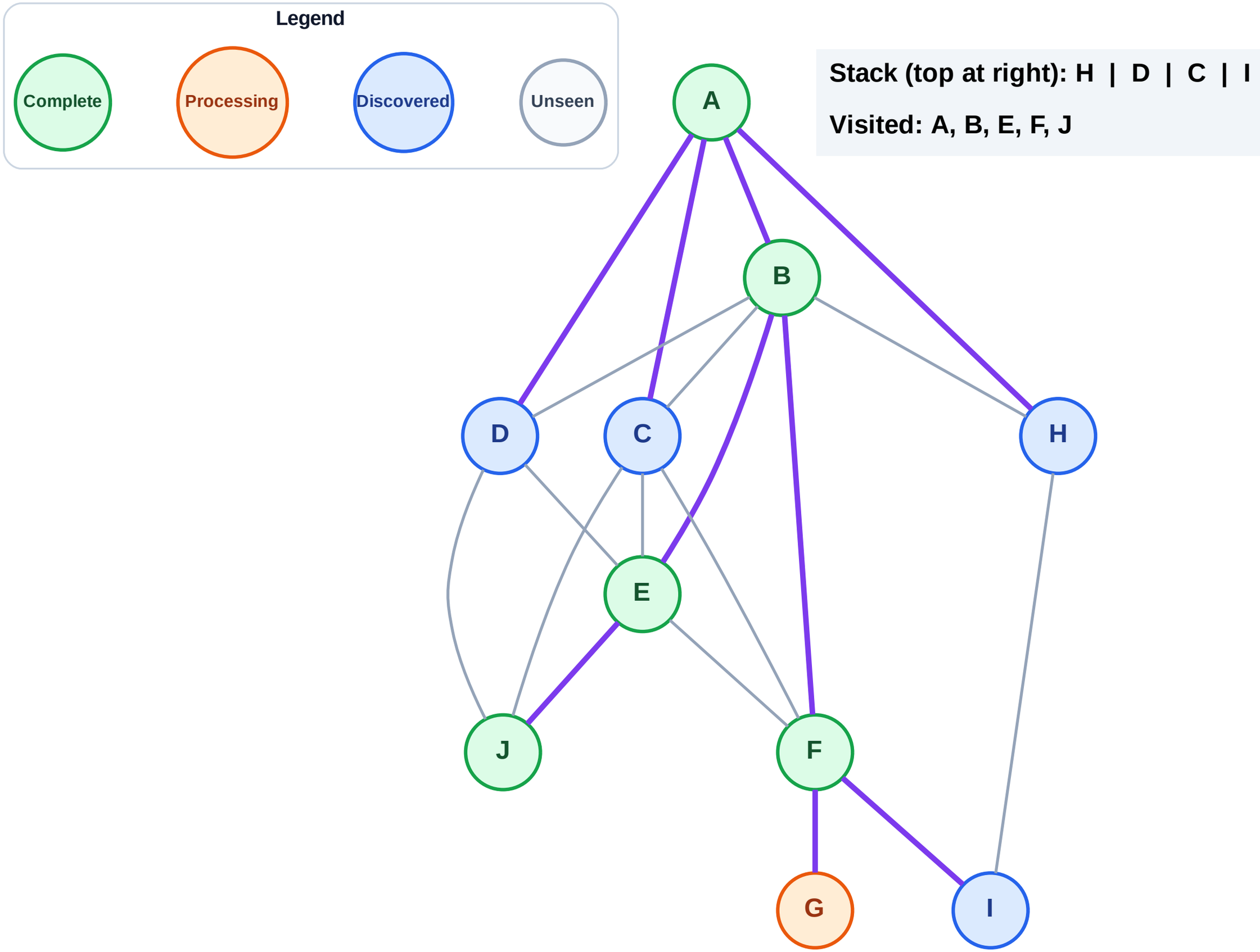
Stack (top at right): H | D | C | I | G

Visited: A, B, E, F, J



DFS Traversal

Step 12: Process node G

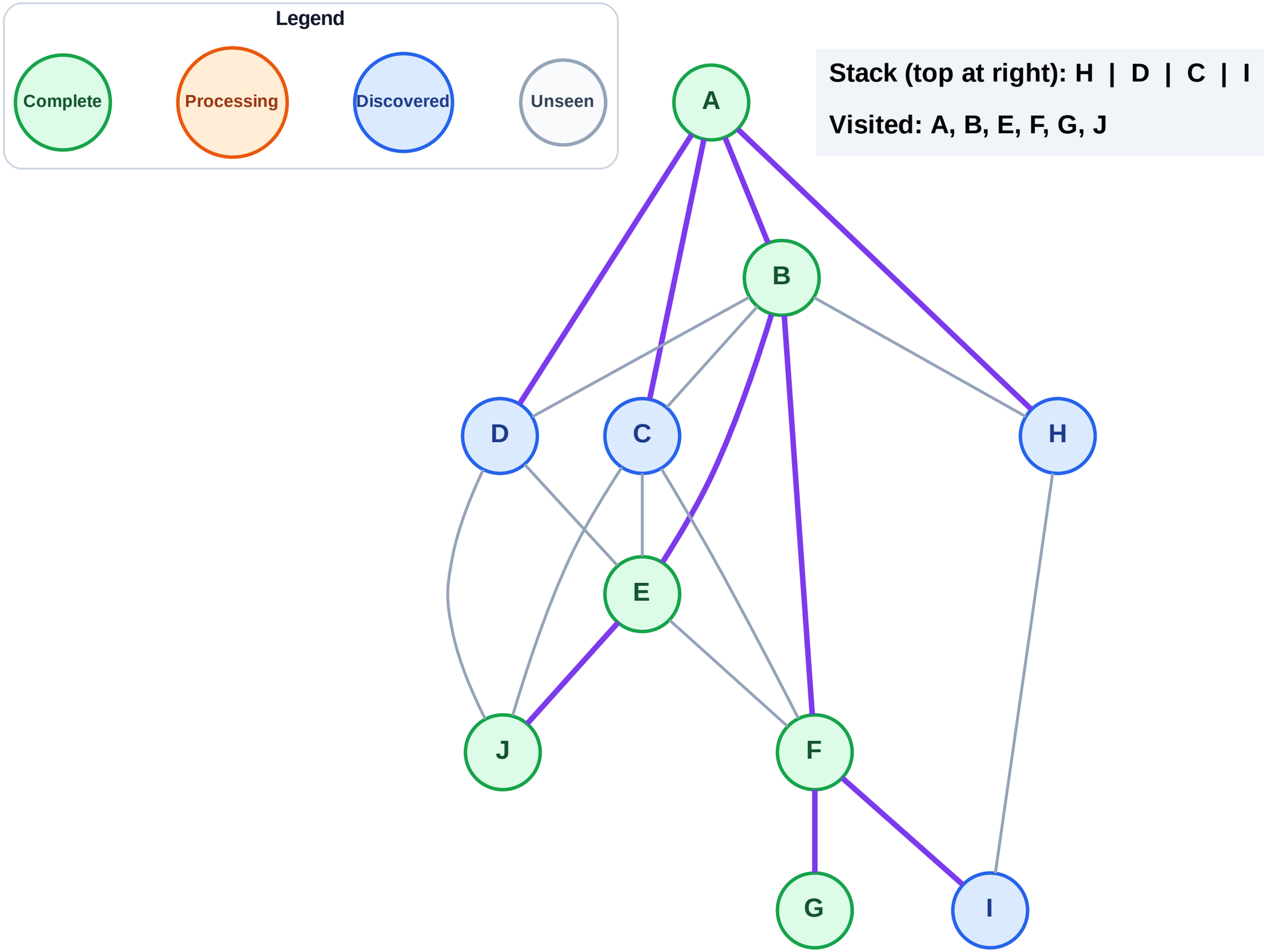


Stack (top at right): H | D | C | I

Visited: A, B, E, F, J

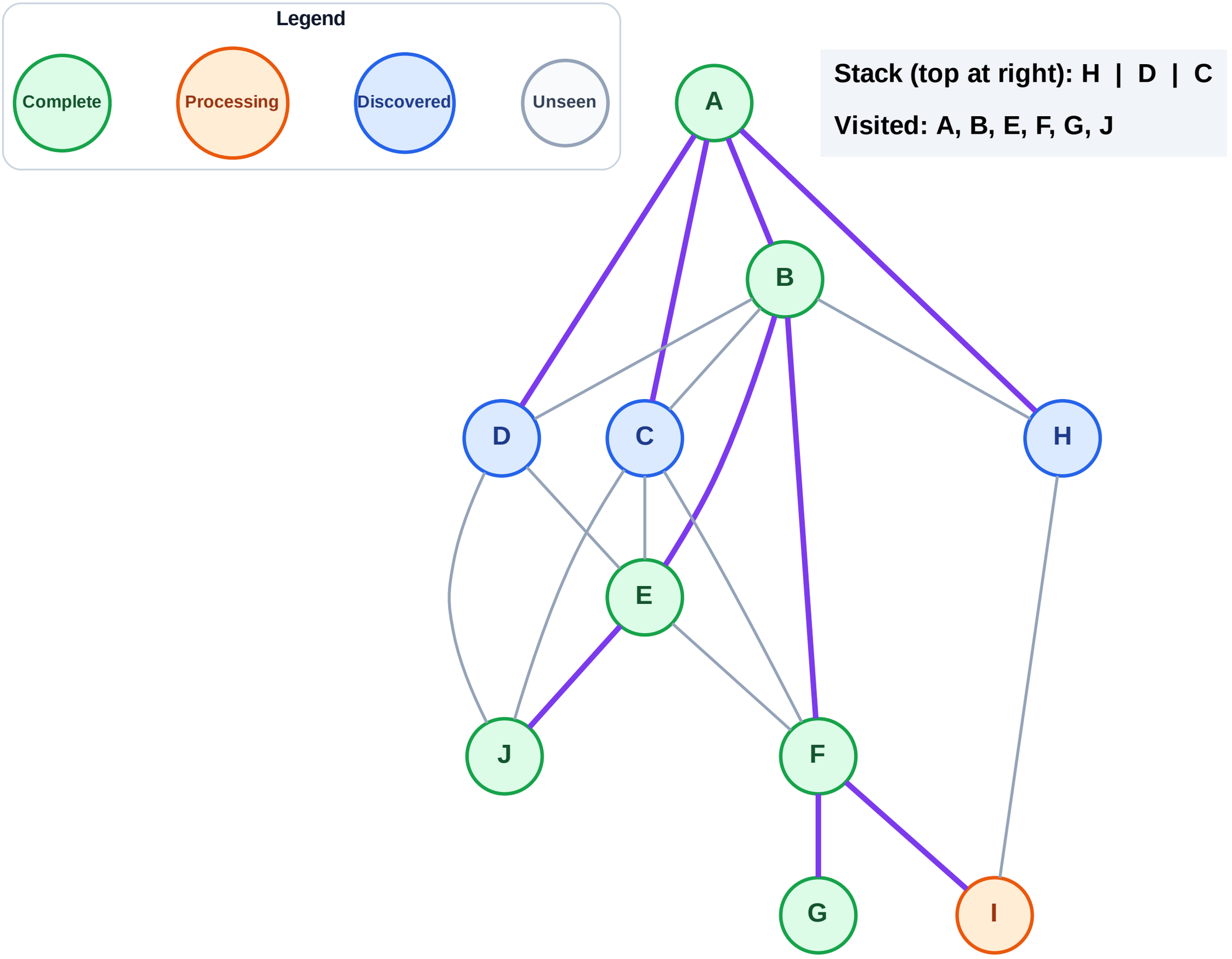
DFS Traversal

Step 13: No new neighbors from G



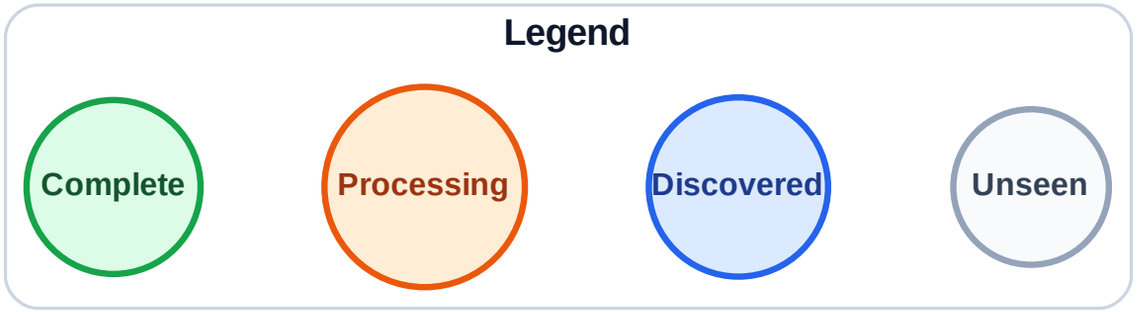
DFS Traversal

Step 14: Process node I



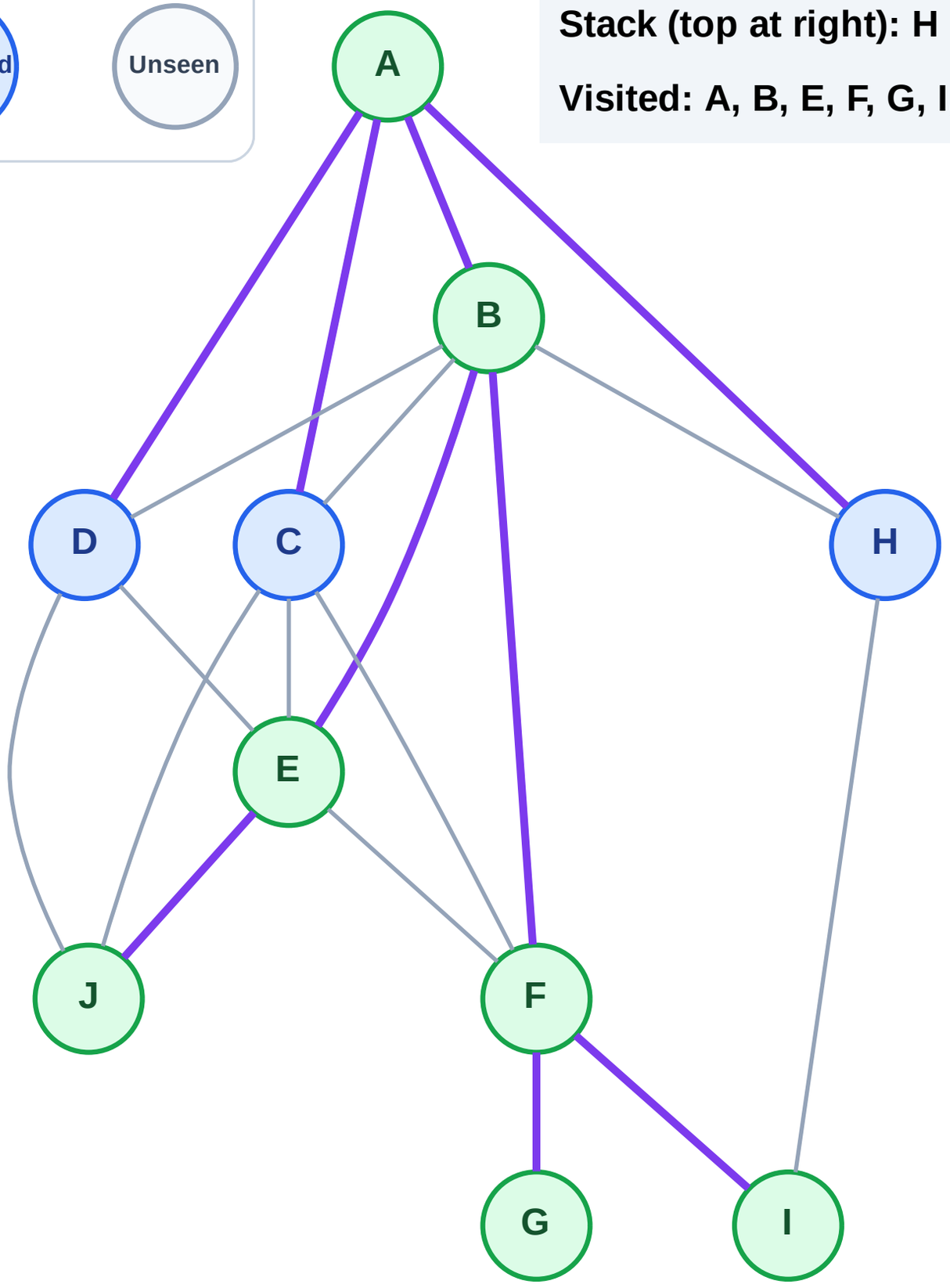
DFS Traversal

Step 15: No new neighbors from I



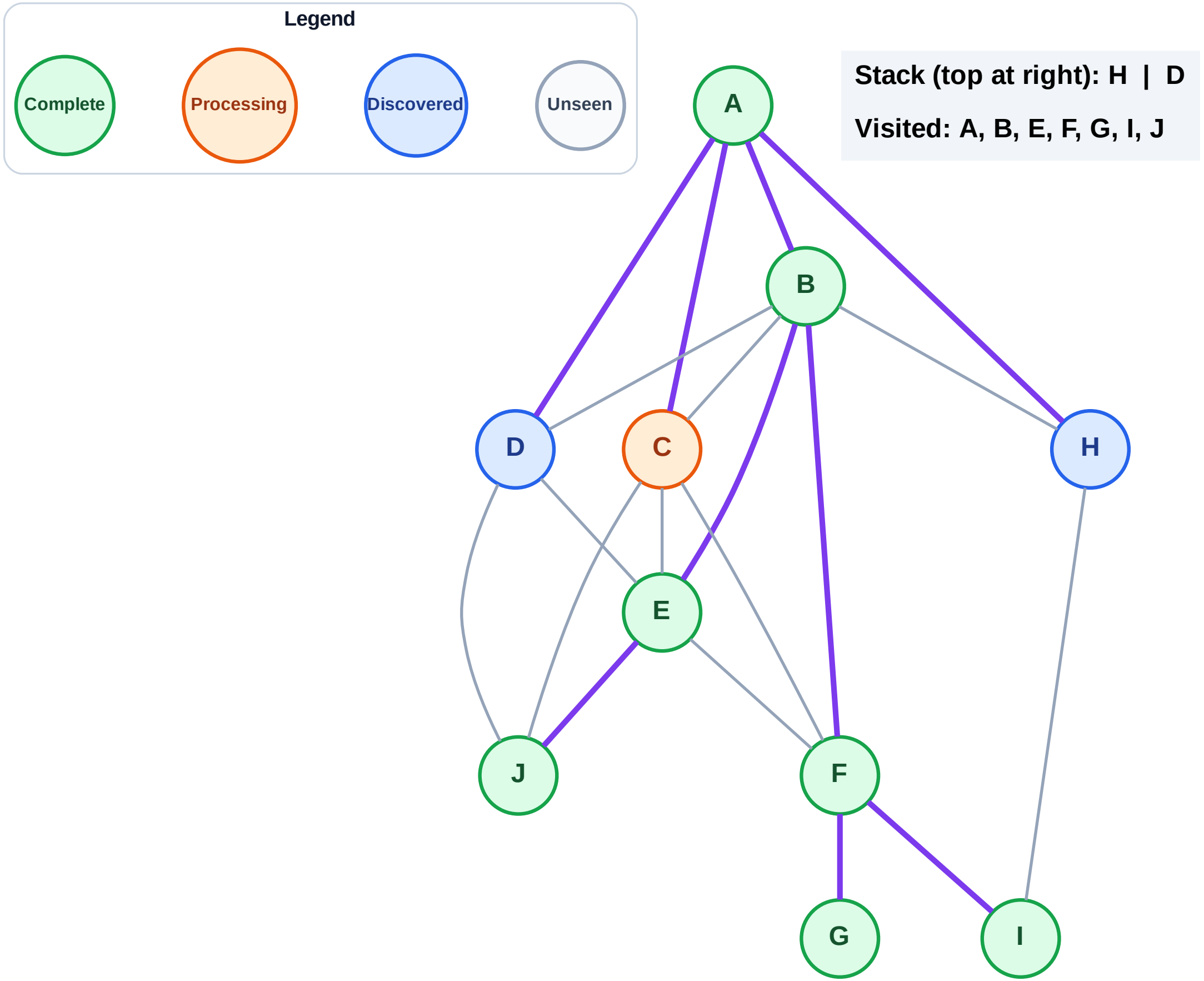
Stack (top at right): H | D | C

Visited: A, B, E, F, G, I, J



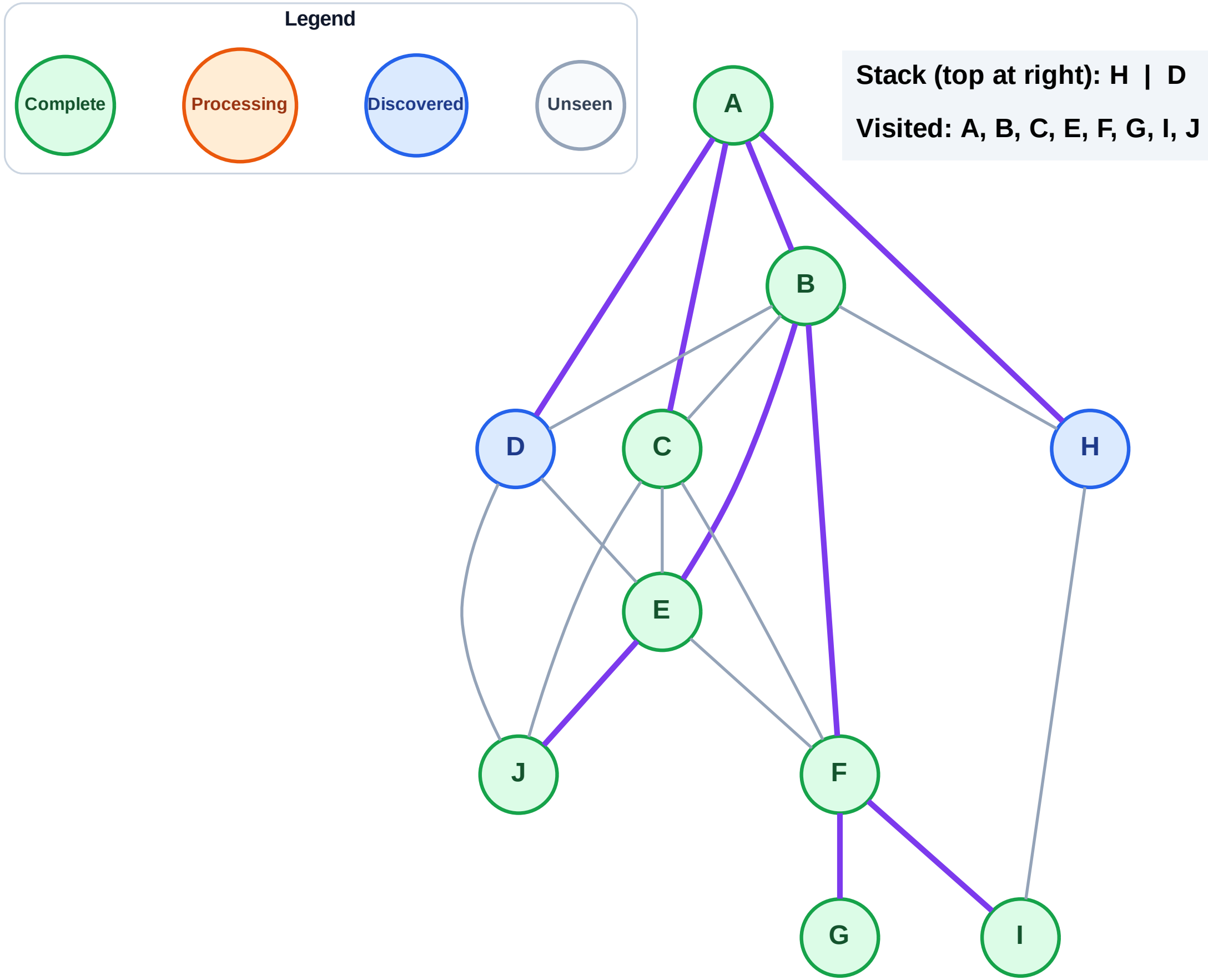
DFS Traversal

Step 16: Process node C



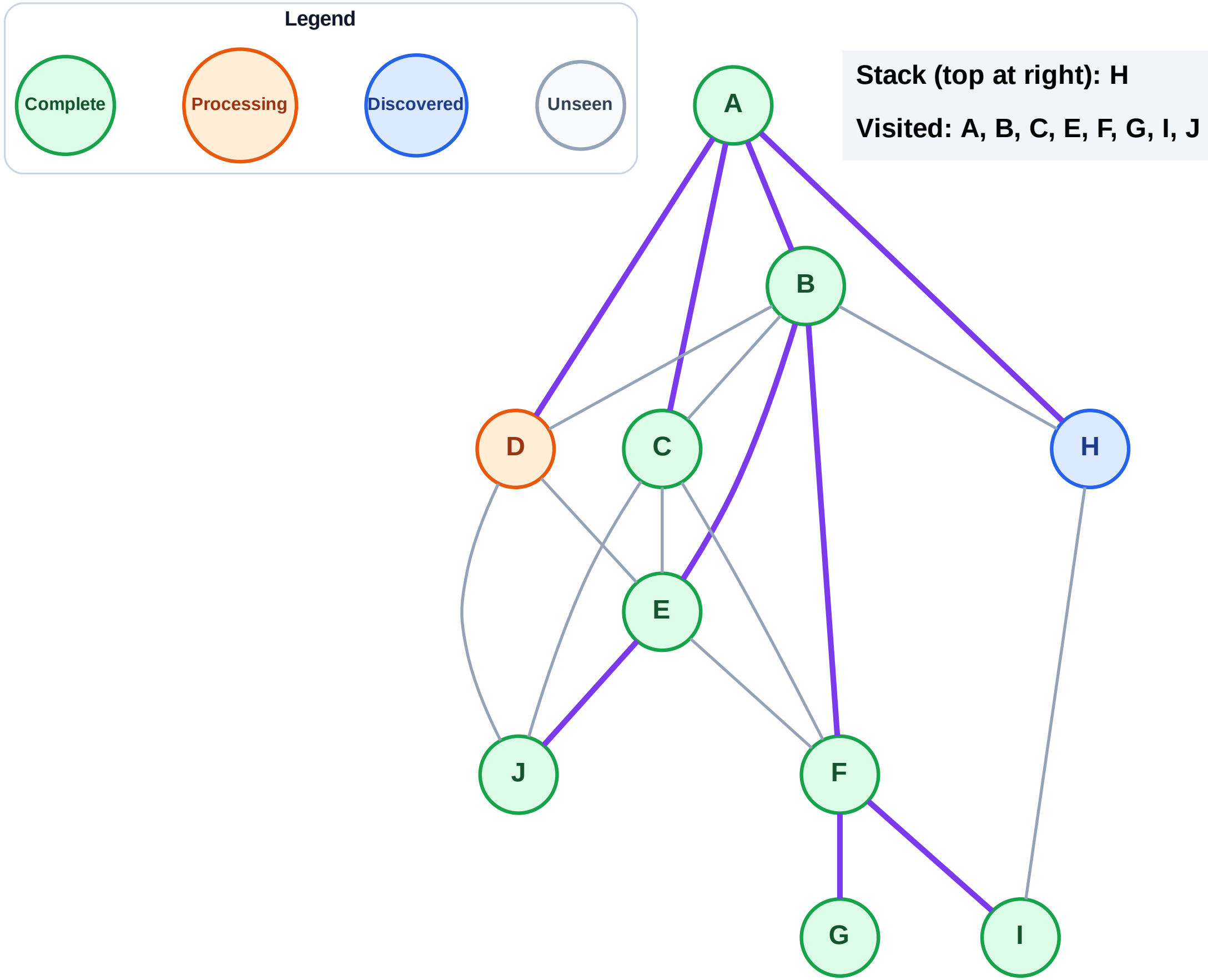
DFS Traversal

Step 17: No new neighbors from C



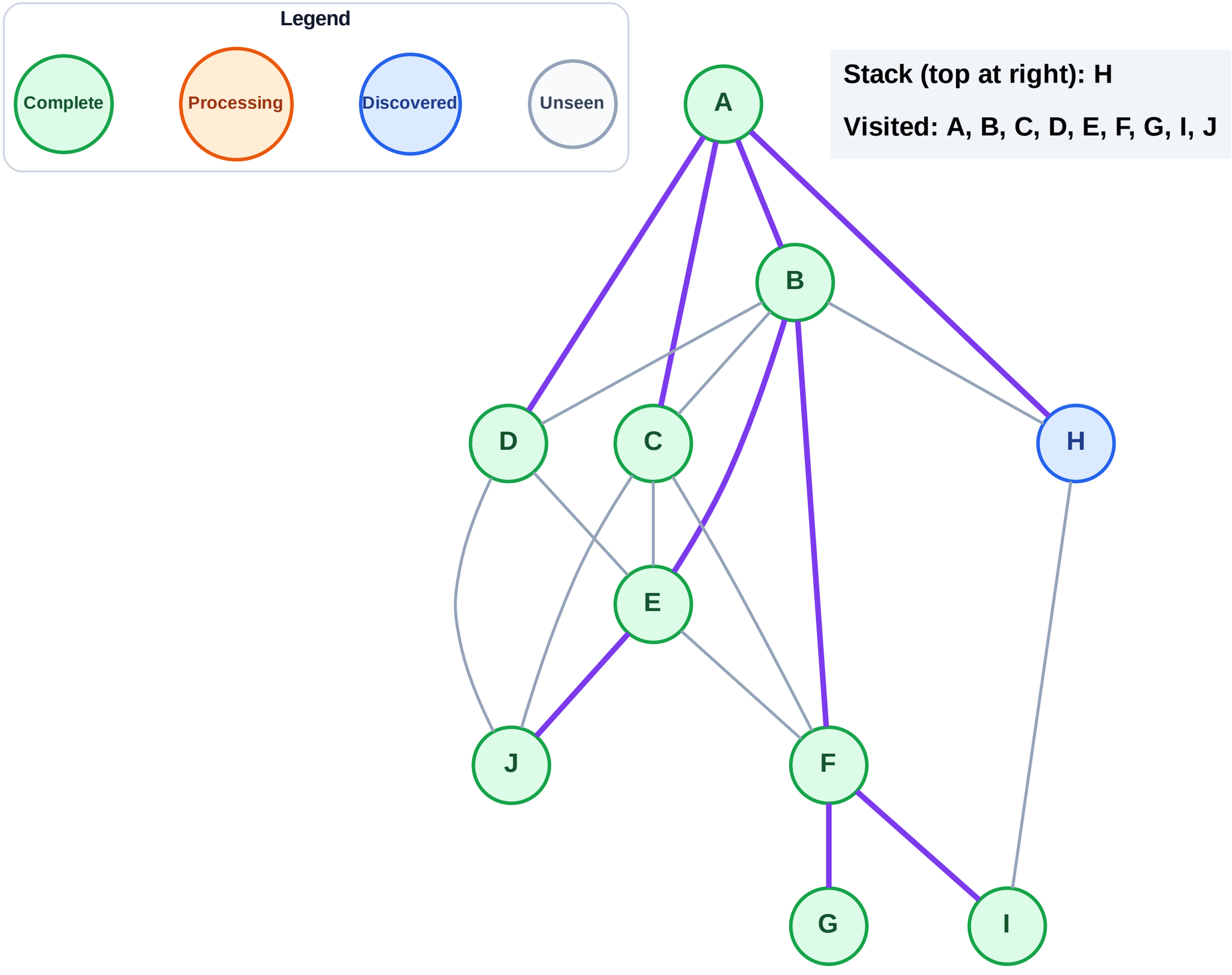
DFS Traversal

Step 18: Process node D



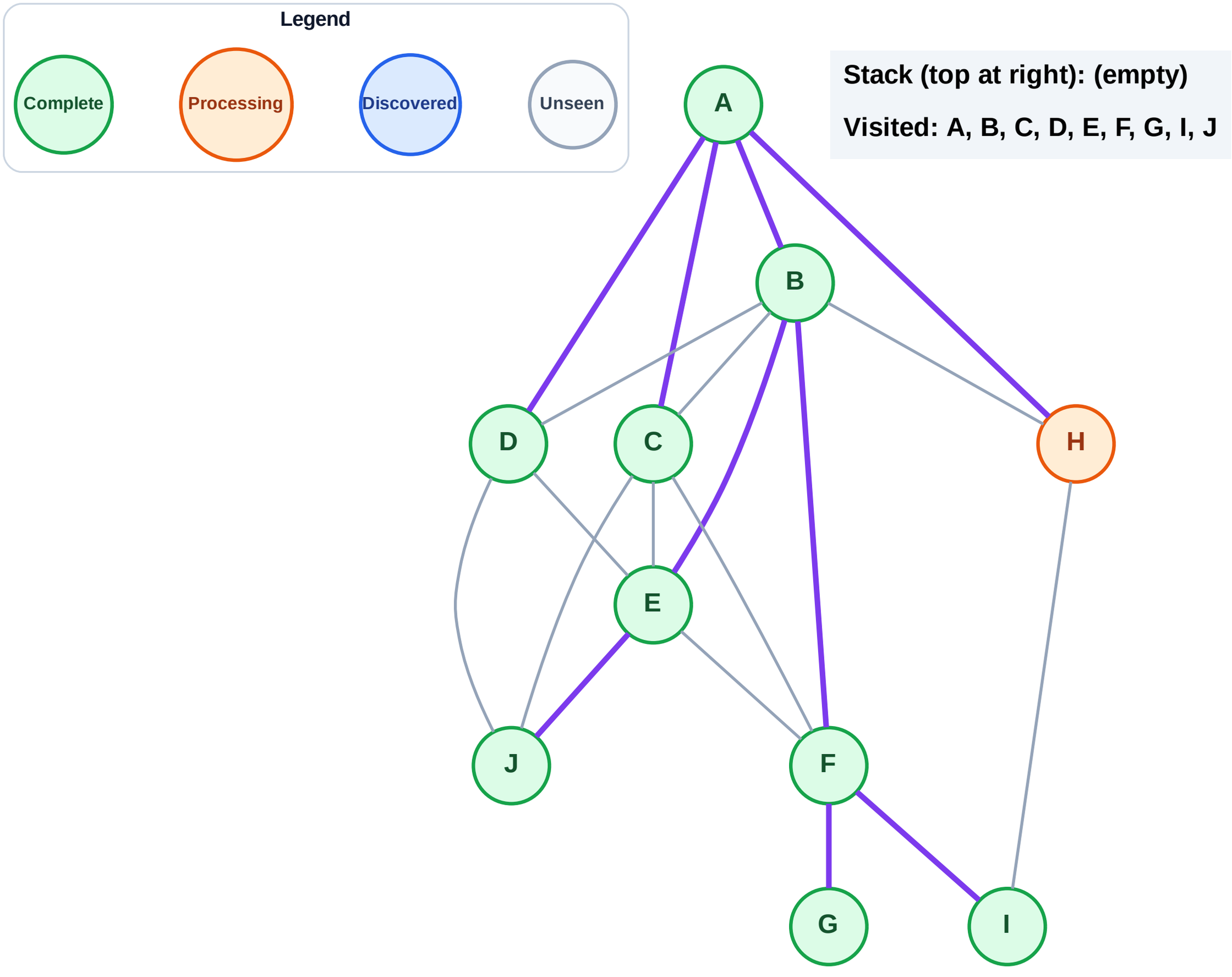
DFS Traversal

Step 19: No new neighbors from D



DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H

